

The Value of a Hospital Obstetric Unit: Access, Costs, and Health*

Thomas Yu[†]

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Abstract

I study *OB provision*, hospitals' decisions to operate obstetric (OB) units. OB provision rates are lower among rural hospitals than urban hospitals and declining overall. How does OB provision affect access to care, the costs of delivering care, and maternal and infant health in equilibrium? Using births and medical claims data, I estimate a model of the hospital OB care market in Arkansas. I find that OB provision benefits mothers substantially by giving them access to more options and shortening travel distances, while effects on health and variable costs are mixed and small. The average OB unit generates a positive externality worth \$102 per Arkansas birth, mostly driven by access to care. I evaluate an Arkansas policy that increases Medicaid prices to encourage OB provision. As OB care becomes less profitable, the policy prevents OB unit closures, returning \$1.1 to \$1.7 in benefits per dollar of spending.

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[†]Yale University, t.yu@yale.edu

1 Introduction

Obstetric (OB) care, or medical care related to pregnancy and childbirth, is largely delivered in hospital OB units. Mothers in rural parts of the United States must often travel great distances to reach hospitals with OB units. The average distance to the nearest OB hospital is 17.3 miles for birthing mothers living in rural counties, as opposed to 7.3 miles for those living in urban counties (Fontenot et al., 2024).

Much of this rural-urban difference is driven by the fact that, even if a rural mother has access to a hospital, that hospital may not have an OB unit. I study *OB provision*: hospitals' decisions to operate OB units and participate in the hospital OB care market. In 2010, 57% of hospitals in rural counties had OB units, as opposed to 70% of hospitals in urban counties; by 2022, these figures dropped to 48% and 64%, respectively (Kozhimannil et al., 2024). In other words, OB provision rates are lower among rural hospitals than urban hospitals by an increasingly wide margin, and they are declining due to OB unit closures. These trends raise concerns about geographic disparities and access to care.

Policymakers have responded by encouraging OB provision. For example, Arkansas's 2025 Healthy Moms, Healthy Babies Act raised the prices paid by Medicaid for OB care by 70% with the goal of preventing OB unit closures (Arkansas Governor, 2025). The value of such a policy depends on the social surplus generated by OB units and, in particular, the extent to which that social surplus exceeds the private surplus they earn for the hospitals operating them.

In this paper, I answer the following questions. How does OB provision affect access to care, the costs of delivering care, and maternal and infant health in equilibrium? What is the net externality of OB provision? What are the consequences of a policy that increases Medicaid prices to encourage OB provision? Focusing on the hospital OB care market in Arkansas in 2022, I quantify the effects of OB provision and evaluate the costs and benefits of the Healthy Moms, Healthy Babies Act.

In a simple illustrative example, I identify four possible welfare effects of OB provision. First, the *access-to-care effect* (giving mothers access to an additional option) increases social welfare. Second, the *fixed cost effect* (fixed cost paid by the OB hospital) decreases social welfare. Third, the *variable cost effect* (variable costs paid by the OB hospital instead of its competitors) may increase or decrease welfare depending on the OB hospital's marginal cost relative to its competitors. Fourth, the *maternal and infant health effect* (changes in mothers' welfare from health) may increase or decrease welfare depending on the OB hospital's health-based quality relative to its competitors. The net externality of OB provision may be positive or negative, so an unregulated market may have too little or too much OB provision relative to the welfare-maximizing outcome.

For my empirical analysis, I combine two datasets from the Arkansas All-Payer Claims Database. The first contains characteristics of all Arkansas births in 2022, including maternal and infant health. The second contains information from medical claims for commercially-insured and Medicaid-insured Arkansas residents in 2022. I supplement these datasets with American Hospital Association data on hospitals' characteristics, Centers for Medicare & Medicaid Services data on hospital quality, regional wages, and supplemental payments, and Census Bureau data on demographics.

I begin with two sets of descriptive facts. First, OB hospital choice is sensitive to travel distance: about half of mothers choose the nearest OB hospital in their choice sets, and this pattern holds across different types of mothers. Second, using variation in distance to a mother's nearest option and the identity of that option, I find suggestive evidence that travel distance is not an important determinant of maternal and infant health while hospital-specific health-based quality is. Together, these facts suggest that OB provision benefits mothers by shortening travel distances, but whether it improves or worsens health depends on variation in health-based quality rather than proximity.

Then, I estimate a model of the hospital OB care market. In the baseline model, commercial prices are set by bargaining between commercial insurers and hospital systems, mothers choose OB hospitals to receive care from, and maternal and infant health is determined. I use a pure characteristics model of OB hospital demand, which has two advantages: it avoids overstating the value of larger choice sets, and it allows me to account for selection by explicitly modeling correlations between unobserved preferences and unobserved underlying health.

I find that mothers strongly prefer nearby hospitals, with a median value equivalent to \$226 more out-of-pocket spending per mile of travel distance. However, the health effect of distance is small: one more mile lowers the probability of experiencing no maternal or infant adverse outcomes by 0.02 percentage points. In a cross-hospital comparison, I find that health-based quality explains only 3% of variation in demand-based quality, suggesting that mothers do not fully internalize health effects when choosing OB hospitals. This motivates me to consider maternal and infant health as a separate component of social welfare not captured by mothers' demand-based welfare.

Using a Nash-in-Nash bargaining framework, I find that hospitals earn large markups: commercial births have average marginal costs of \$4,900 and markups of \$3,600, and Medicaid births have average marginal costs of \$1,200 (adjusted downward for supplemental payments) and markups of \$1,800.

I use the estimated model to quantify the effects of OB provision for each OB hospital by simulating equilibrium outcomes if its OB unit closed. On average, an OB hospital's access-to-care effect is worth \$159 per Arkansas birth, reflecting the high value that mothers place

on access to more options and shorter travel distances. Variable cost effects (on average, \$5 per Arkansas birth) and maternal and infant health effects (on average, \$7 per Arkansas birth) are mixed and much smaller: some hospitals reduce variable costs or improve health by operating OB units, while others increase variable costs or worsen health by doing so. OB provision benefits rural mothers more than urban mothers: the average OB hospital improves mothers' demand-based welfare by \$214 per rural mother and \$128 per urban mother. Then, I compute the net externality of OB provision by subtracting the private surplus internalized by hospital systems. On average, an OB hospital earns \$2.2 million in variable profit for its system and generates an additional externality of \$3.3 million, or \$102 per Arkansas birth. These externalities are mostly driven by access to care.

I extend the model to include hospitals' OB provision decisions and estimate the fixed cost of OB provision. I use revealed-preference bounds to set-identify expected fixed cost, and I find that it is between \$0.56 million and \$2.23 million. As an out-of-sample check, I verify that the four hospitals that actually closed their OB units in the two years after the sample period are among the six OB hospitals that the model predicts to be the least profitable.

Finally, I evaluate the Healthy Moms, Healthy Babies Act's 70% increase in Medicaid prices. I simulate realistic conditions that may induce OB unit closures: 10%, 20%, and 30% declines in variable profits. Without the policy, 27 to 36 of the 37 OB hospitals remain. The policy prevents most of these closures and increases social welfare by \$3 to \$21 million, largely through access to care and with larger benefits for rural mothers than urban mothers. The policy's marginal value of public funds is 1.1 to 1.7: each dollar of Medicaid spending is a dollar transferred to hospitals plus \$0.1 to \$0.7 in additional social welfare.

Related Literature First, this paper is related to the literature on effects of hospital and OB unit closures. Some papers use demand and supply models to quantify welfare effects of closures. For example, Capps et al. (2010) study five hospital closures and find increases in social welfare. Other papers use difference-in-differences strategies to quantify health effects of closures. For example, Joynt et al. (2015) and Petek (2022) find zero mortality effects of hospital closures. Most closely related to this paper, Battaglia (2025) and Fischer et al. (2024) find that OB unit closures have negligible to slightly positive maternal and infant health effects. These papers are informative about the effects of market participation, but only among hospitals that experienced closures and, for those using difference-in-differences, only on average. I estimate hospital-specific effects of OB provision among all OB hospitals, and I combine health with other components of welfare. I also explicitly model OB provision decisions, allowing me to evaluate a policy that prevents OB unit closures.

Second, this paper is related to the literature on optimal firm entry. A well-established

theoretical literature shows that an unregulated market may have too few or too many participating firms relative to the welfare-maximizing outcome (Mankiw & Whinston, 1986; Spence, 1976). Some empirical papers quantify the welfare effects of market participation in real-world markets; for example, Berry et al. (2016) find too many participants in the local radio market. I consider a new market (hospital OB care), include health as another component of welfare affected by market participation, and find a positive net externality of market participation.

Third, this paper is related to the literature on health care provider quality estimation. A common issue when estimating provider quality is selection: a patient's outcomes are only observed at the provider they choose, and provider choice may be correlated with unobserved underlying health. A popular strategy in this literature is to use patients' home locations as exogenous choice shifters (Einav et al., 2025; Geweke et al., 2003; Olenski & Sacher, 2024). In my model, I use choice shifters based on not only home locations, but also out-of-pocket spending and insurance networks. Furthermore, I consider new applications of provider quality estimates: quantifying the effects of providers' market participation and evaluating a policy that affects market participation among providers of varying quality.

Fourth, this paper is related to the literature on models of hospital markets. My model is most similar to that of Gowrisankaran et al. (2015) in which insurers and networks are taken as exogenous and commercial prices are set by bargaining between insurers and hospital systems. I adapt the model to one service line, hospital OB care, and extend it by adding maternal and infant health and OB provision decisions.

Fifth, this paper is related to the literature on pure characteristics demand models. In my model of OB hospital demand, I omit an unobserved consumer-product-level utility shock that often appears in discrete choice models. One reason to do so is that demand models with consumer-product-level utility shocks tend to overstate the value of larger choice sets, which is central to my research question (Akerberg & Rysman, 2005; Berry & Pakes, 2007). Another reason is that a pure characteristics model makes it feasible to allow all unobserved preferences to correlate with unobserved underlying health, capturing any possible selection. I use methods from Cuesta and Tebaldi (2025), who show that rich individual-level data can help overcome a pure characteristics model's computational challenges.

Sixth, this paper is related to the literature on fixed cost estimation. Because firms that participate in a market may have lower draws of unobserved fixed cost shocks than non-participants, fixed cost estimation cannot rely on expectations conditional on participation. I follow the strategy of Eizenberg (2014), who assumes worst-case fixed cost bounds to construct expectations unconditional on participation. I apply this strategy to estimate the fixed cost of OB provision, and I use the estimate to evaluate policy.

Outline In Section 2, I describe features of the empirical setting, Arkansas in 2022, including the data I use and descriptive facts. In Section 3, I present a simple example that illustrates the effects of OB provision. In Section 4, I develop an empirical model of demand, health, and prices in the hospital OB care market and discuss identification, estimation, and results. In Section 5, I use the estimated model to quantify the effects of OB provision. In Section 6, I extend the model to include hospitals' OB provision decisions and estimate the fixed cost of OB provision. In Section 7, I evaluate Arkansas's 2025 Healthy Moms, Healthy Babies Act, a policy that raised Medicaid prices to encourage OB provision. In Section 8, I conclude.

2 Arkansas in 2022: Setting, Data, and Descriptive Facts

2.1 Setting

The empirical setting of this paper is Arkansas in 2022. Arkansas is highly rural: among its population of 3 million people, 45% live in rural areas, more than double the national average (Census Bureau, 2024). Arkansas also faces significant maternal and infant health challenges. Its maternal mortality rate (0.35 per thousand births) is 90% higher than the national average, and its infant mortality rate (8.2 per thousand births) is 46% higher than the national average (March of Dimes, 2025). Thus, health effects are especially salient.

I exclude three counties because they are parts of metropolitan areas with population centers in other states: Crittenden County (part of the Memphis metropolitan area) and Miller and Little River Counties (parts of the Texarkana metropolitan area). I exclude specialty hospitals (e.g., psychiatric, rehabilitation, or long-term acute care hospitals) because they do not provide OB care. I exclude hospitals in the Veterans Affairs (VA) system because they serve a particular population that is mostly missing from the claims data I use.

The setting contains 71 hospitals, 37 with OB units and 34 without OB units. Figure 1 shows the locations of these hospitals along with population density. Although Arkansas's overall density is low, it has a few relatively dense urban areas: most notably, the Central Arkansas (Little Rock) and Northwest Arkansas (Fayetteville–Springdale–Rogers) metropolitan areas. Urban areas often have multiple clustered OB hospitals, while rural areas often only have one or zero.

Of the 37 OB hospitals, 9 have neonatal intensive care units (NICUs). One OB hospital, the University of Arkansas for Medical Sciences (UAMS), has an obstetrics and gynecology residency program, and another, Willow Creek Women's Hospital (WCWH), is dedicated to women's health care. These hospitals may have particular demand-based quality, health-based quality, or marginal costs. Of the 71 hospitals, 37 belong to multi-hospital systems. Together,

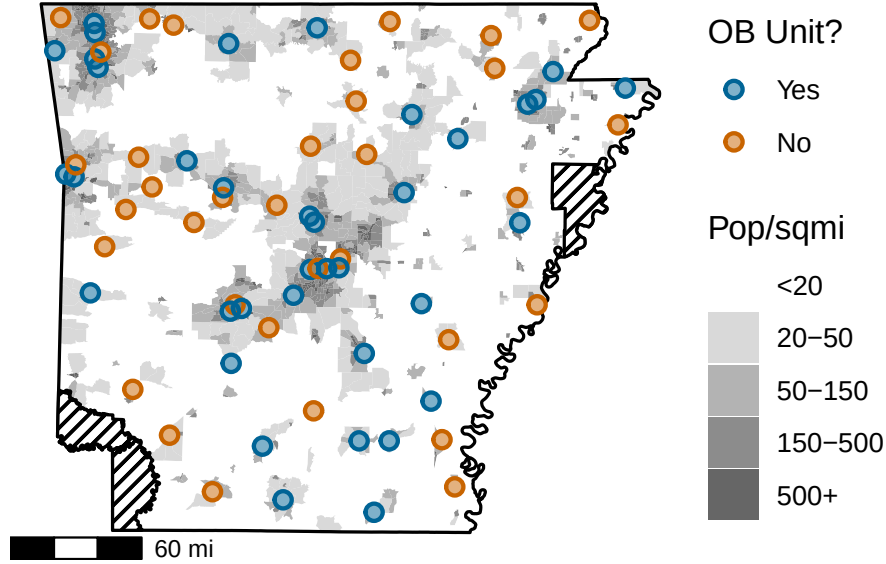


Figure 1: OB Provision in Arkansas in 2022

Data sources: AHA, Census. Sample: hospitals in Arkansas in 2022 (except Crittenden/Miller/Little River Counties and specialty/VA hospitals). Pop densities by census block group. Hatched areas are excluded counties.

they are owned by 9 multi-hospital systems. These hospitals’ decisions about OB provision and commercial price-setting are made jointly at the system level.

In 2025, Arkansas policymakers passed the Healthy Moms, Healthy Babies Act, which invests \$45 million per year on measures to improve maternal and infant health; more than half of the investment, \$26 million, is dedicated to raising the prices paid by Medicaid for OB care by 70% (Arkansas Governor, 2025). The motivation for this increase is to prevent OB unit closures, or encourage OB provision, by making it more profitable to deliver OB care. In Section 7, I evaluate the costs and benefits of this Medicaid price increase.

2.2 Data

I use data from the Arkansas All-Payer Claims Database, as provided by the Arkansas Center for Health Improvement. I combine two datasets. The first contains characteristics of all Arkansas births in 2022, including mothers’ characteristics, chosen OB hospitals, and maternal and infant health. The second contains information from medical claims for commercially-insured and Medicaid-insured Arkansas residents in 2022, including payments, out-of-pocket spending, insurers, and networks. In Appendix A, I detail how the main data are constructed.

I use six additional sources of data. First, the 2022 Annual Survey from the American Hospital Association (AHA) provides data on hospitals’ characteristics, including for-profit

status, beds, and NICUs. Second, the 2022 Provider Data Catalog from the Centers for Medicare & Medicaid Services (CMS) provides data on hospital quality, including reviews from Medicare patient surveys, Medicare mortality rates, and Medicare readmission rates. Third, the 2022 CMS Wage Index provides a measure of regional hospital worker wages. Fourth, CMS's Medicaid Budget and Expenditure System provides data on hospitals' supplemental payments for treating Medicaid patients. Fifth, the Census Bureau's 2018–2022 American Community Survey provides data on regional demographics. Sixth, the Census Bureau's 2020 Delineation Files define counties as urban (metropolitan) or rural (not metropolitan).

There are 32,307 births in the sample, and their summary statistics are shown in Table 1. A little over half are commercial, almost all of the rest are Medicaid, and a few are neither (e.g., uninsured). Commercial and Medicaid births have similar average distances traveled and adverse outcome rates, where adverse outcomes are either maternal (perineal laceration, blood transfusion, unplanned hysterectomy, or uterine rupture) or infant (5-minute Apgar score less than 4, death, injury, or seizure).

I match 53% of commercial births and 79% of Medicaid births to claims data. There are two reasons why commercial and Medicaid births may be missing claims data. First, some commercial insurers are not required to report to the All-Payer Claims Database: commercial employer-sponsored self-insured plans and plans with fewer than 2,000 enrollees. Second, there may be ambiguity in mothers' names or other data collection errors.

I use births matched to claims data as the main estimation sample. When I compute aggregate measures (e.g., variable costs or profits), I assume commercial births with claims data are similar to those missing claims data, and Medicaid births with claims data are similar to those missing claims data. In other words, I re-weight commercial and Medicaid births to account for their rates of missing claims data. I include the few births covered by neither commercial insurance nor Medicaid in the Medicaid category.¹

In terms of payments, commercial births have higher payments (\$8,400 on average) with a substantial portion paid by mothers out-of-pocket (\$1,200 on average). Medicaid births have lower payments (\$3,000 on average) and zero out-of-pocket spending.

In Appendix B, I show birth summary statistics split by whether they match to claims data, birth summary statistics by quarter of the year, rates of different adverse outcomes, and hospital-level summary statistics.

I highlight two notes about the data. First, the commercial category includes Medicaid-funded plans. The Arkansas Health and Opportunity for Me (ARHOME) program uses Medicaid funds to buy commercial insurance for residents with incomes between the pre-expansion

¹For many of these births, either the mother is uninsured and would qualify for Medicaid under the generous eligibility rules for pregnant women, or she is covered by another public program (e.g., Medicare or VA).

Table 1: Birth Summary Statistics

	Commercial	Medicaid	Other
Mean			
Distance (mi)	16.6	16.8	13.8
Any Adverse Outcome	0.03	0.03	0.03
Matched to Claims	0.53	0.79	0
Payment (\$)	8399	2962	—
Out-of-pocket (\$)	1180	0	—
ARHOME	0.50	0	0
Rural	0.29	0.38	0.28
Age (yr)	28.6	26.0	28.7
BMI	28.8	29.0	27.3
Prior Births>0	0.57	0.67	0.69
Prior C-sections>0	0.14	0.16	0.09
Prenatal Visits	11.0	9.3	9.0
Gestational Age (wk)	38.2	38.1	38.8
Planned C-section	0.22	0.22	0.13
Outside Option	0.01	0.01	0.33
UAMS (Teaching)	0.07	0.13	0.05
WCWH (Women's)	0.05	0.07	0.15
Public	0.08	0.16	0.07
Non-profit	0.74	0.63	0.40
For-profit	0.17	0.20	0.21
NICU	0.57	0.51	0.47
Non-OB Beds	301.8	251.5	155.1
Observations	18003	13360	944

Adverse outcomes: maternal (perineal laceration, blood transfusion, unplanned hysterectomy, uterine rupture) or infant (Apgar<4, death, injury, seizure). Additional summary statistics in [Appendix B](#).

and expanded Medicaid thresholds. ARHOME plans feature zero out-of-pocket spending like traditional Medicaid, but they feature limited networks and negotiated prices like commercial insurance. In this paper, I classify ARHOME plans as commercial. 50% of the commercial births matched to claims are associated with ARHOME.

Second, distances between mothers and hospitals are imputed expected distances. I observe mothers' home locations at the county level. From the Census Bureau data, I observe populations of women by age bin at the census block group level. I measure the haversine distances between the centroids of census block groups and hospitals. Finally, I use each mother's county and age to construct her probability of living in each census block group and impute her expected distances to hospitals.

2.3 Descriptive Facts

I present two sets of descriptive facts that illustrate important features of this setting. First, I show that OB hospital choice is sensitive to travel distance. Figure 2 shows how often mothers choose their nearest, second-nearest, third-nearest, fourth-nearest, and fifth-nearest options in their choice sets, where choice sets are defined as all in-network OB hospitals. About half of mothers choose their nearest option, and this pattern holds across different types of mothers. This suggests that travel distance is an important driver of OB hospital choice. It also points to a channel through which OB provision improves mothers' welfare: an additional OB hospital gives some mothers a shorter distance to their nearest options, which, as revealed by their choices, they highly value.

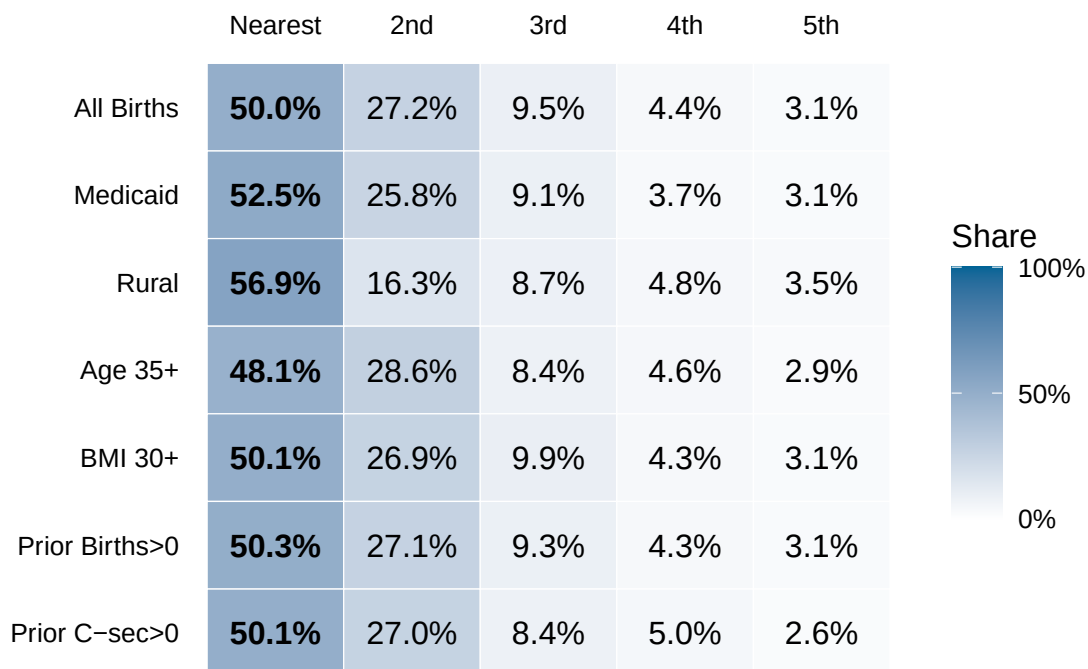


Figure 2: Share of Mothers Who Choose Their Nearest Options

Sample: births with claims data, excluding outside option births. Choice set: in-network OB hospitals.

Second, I present evidence about maternal and infant health, suggesting that travel distance is not an important determinant of health while hospital-specific health-based quality is. Let Healthy_i be an indicator for whether mother i experiences no maternal or infant adverse

outcomes. I estimate the following equation:

$$\text{Healthy}_i = \mathbb{1} \left(\tau' \mathbf{x}_i + \tau^{\text{Dist}} \text{DistNearest}_i + \sum_{j \in J_{m(i)}^{\text{OB}}} \phi_j (j \text{ is nearest to } i) + \omega_i > 0 \right), \quad \omega_i \sim N(0, 1) \quad (1)$$

where $\mathbf{x}_i$ consist of a constant, mother's age, mother's race, mother's BMI, a diabetes indicator, a hypertension indicator, a smoker indicator, a prior birth indicator, a prior caesarean section indicator, logged home county population density, and home county population shares by income bin. DistNearest_i is mother i 's distance to the nearest OB hospital in her choice set, and $J_{m(i)}^{\text{OB}}$ is her choice set. ω_i is unobserved underlying health. ϕ_j is a hospital-specific nearest-option health effect, and I estimate it as a random effect with $\phi_j \sim N(0, \sigma_\phi^2)$.

One parameter of interest is τ^{Dist} , which captures the health effect of distance to the nearest available OB hospital. Another parameter of interest is σ_ϕ , which captures the spread of nearest-option health effects ϕ_j across OB hospitals.

This exercise relies on two key assumptions: mothers tend to choose their nearest option (which is supported by Figure 2), and mothers' home locations are unrelated to their unobserved underlying health, conditional on observed characteristics.² I estimate Equation 1 with maximum likelihood estimation, and Table 2 shows the results. I convert the estimates to average marginal effects, where the nearest-option health effects are relative to a reference hospital, UAMS.

Table 2: Maternal and Infant Health Descriptive Results

	Avg $\Delta\text{Pr}(\text{Healthy})$ (%)
Distance to Nearest Option	
Per Mile	0.013 (0.019)
Per Standard Deviation (8.6 mi)	0.115 (0.159)
Nearest-Option Effects	
Standard Deviation Across Hospitals	0.566 (0.218)
Observations	19843

Results from estimating Equation 1. Sample: births with claims data, excluding outside option births. Drop hospitals that are nobody's nearest option (2 hospitals). Average marginal effects. Reference hospital for nearest-option effects: UAMS. Model-based SEs.

The health effect of distance to the nearest available OB hospital, whether measured per

²In Appendix D, I provide descriptive evidence suggesting that conditional on observed characteristics, mothers' home locations are unrelated to OB care.

mile or per standard deviation, is near zero and statistically insignificant. This is consistent with the findings of Battaglia (2025) and Fischer et al. (2024), who find no differences in health effects across mothers facing different OB-unit-closure-driven increases in distance to the nearest OB hospital. However, there is notable variation in hospital-specific nearest-option effects. The standard deviation of nearest-option effects across hospitals is large, 0.57 percentage points with a standard error of 0.22.

On the one hand, despite the sensitivity of mothers' choices to travel distance, its health effect is small. On the other hand, hospital-specific health-based quality is important. This evidence suggests that whether an OB unit improves or worsens aggregate maternal and infant health depends on its health-based quality relative to its competitors rather than how much it reduces travel distances.

3 Simple Example of OB Provision

In this section, I present a simple example that illustrates the effects of OB provision. There is a continuum of mothers of measure 1 distributed uniformly between two hospitals. At one end, Hospital A operates an OB unit. At the other end, Hospital B may or may not operate an OB unit. An OB unit has fixed cost $FC > 0$. Hospital A's OB unit has constant marginal cost $MC_A > 0$, and Hospital B's OB unit (if it operates one) has constant marginal cost $MC_B > 0$. The price of hospital OB care is $p > \max\{MC_A, MC_B\}$.

Each mother chooses to receive care from her preferred OB hospital, and there is no outside option. Mothers incur costs from price and travel distance.³ Mothers value maternal and infant health, which improves with higher OB hospitals' health-based quality and may also depend on travel distance. I allow for imperfect information by assuming mothers have the following beliefs about health (which may or may not be correct): OB hospitals have the same health-based quality, and travel distance does not affect health. Thus, although mothers are affected by differences in maternal and infant health, they do not internalize these differences when making choices and simply choose their nearest OB hospitals.

At first, suppose only Hospital A operates an OB unit. All mothers choose Hospital A. Because prices are transfers, social welfare consists of aggregate travel costs, aggregate fixed and variable costs, and the value of aggregate maternal and infant health. In other words, social welfare is

$$\text{Welfare}_{A \text{ only}} = -\text{TravelCosts}_{A \text{ only}} - FC - MC_A + \text{Health}_{A \text{ only}} \quad (2)$$

³In reality, mothers do not pay the full price because of insurance. In this simple example, mothers' choices can be interpreted as joint mother-insurers' choices, and mothers' welfare as joint mother-insurers' welfare.

Then, suppose Hospital B operates an OB unit as well. The half of mothers located nearer to Hospital B than to Hospital A switch to choosing Hospital B. Social welfare is

$$\text{Welfare}_{A \text{ and } B} = -\text{TravelCosts}_{A \text{ and } B} - 2FC - (\text{MC}_A + \text{MC}_B)/2 + \text{Health}_{A \text{ and } B} \quad (3)$$

The change in social welfare due to Hospital B's OB provision is

$$\Delta \text{Welfare} = \underbrace{(-\Delta \text{TravelCosts})}_{>0} + \underbrace{(-FC)}_{<0} + \underbrace{(\text{MC}_A - \text{MC}_B)/2}_{>0 \text{ or } <0} + \underbrace{(\Delta \text{Health})}_{>0 \text{ or } <0} \quad (4)$$

where $\Delta \text{TravelCosts}$ is the change in aggregate travel costs and ΔHealth is the value of the change in aggregate maternal and infant health.

Hospital B's OB provision can affect welfare through four channels. First, it can have an *access-to-care effect* (giving mothers access to an additional option that reduces some of their travel costs) which increases social welfare. Second, it can have a *fixed cost effect* (fixed cost paid by Hospital B) which decreases social welfare. Third, it can have a *variable cost effect* (variable costs paid by Hospital B instead of Hospital A) which may increase or decrease social welfare depending on Hospital B's marginal cost relative to Hospital A.

Fourth, it can have a *maternal and infant health effect*. Due to imperfect information, maternal and infant health does not affect mothers' choices, but it may still affect their welfare. Mothers who switch from Hospital A to Hospital B experience two health effects: a potential positive effect due to shorter travel distance if care is time-sensitive, and a positive or negative effect depending on Hospital B's health-based quality relative to Hospital A. Aggregating over mothers, the health effect may increase or decrease social welfare.

Consider the outcome if a social planner decides whether to operate an OB unit at Hospital B to maximize social welfare. The social planner's OB provision decision is

$$\text{Prov}_{\text{Planner}} = \begin{cases} 1 & \text{if } -\Delta \text{TravelCosts} + (\text{MC}_A - \text{MC}_B)/2 + \Delta \text{Health} > FC \\ 0 & \text{if } -\Delta \text{TravelCosts} + (\text{MC}_A - \text{MC}_B)/2 + \Delta \text{Health} < FC \end{cases} \quad (5)$$

Alternatively, consider an unregulated market in which Hospital B decides whether to operate an OB unit to maximize its own profit. Let $\pi_B = (p - \text{MC}_B)/2$ be Hospital B's variable profit upon OB provision. Hospital B's OB provision decision is

$$\text{Prov}_{\text{Private}} = \begin{cases} 1 & \text{if } \pi_B > FC \\ 0 & \text{if } \pi_B < FC \end{cases} \quad (6)$$

On the one hand, the unregulated market has too little OB provision relative to the welfare-

maximizing outcome (Hospital B does not operate an OB unit while the social planner would) if $\pi_B < FC < -\Delta\text{TravelCosts} + (MC_A - MC_B)/2 + \Delta\text{Health}$. On the other hand, it has too much OB provision (Hospital B operates an OB unit while the social planner would not) if $-\Delta\text{TravelCosts} + (MC_A - MC_B)/2 + \Delta\text{Health} < FC < \pi_B$.

Externalities Another way to express welfare effects is in terms of externalities. Out of the change in social welfare, Hospital B internalizes its own total profit. Thus, the net externality generated by Hospital B's OB provision decision is

$$\text{Externality} = \Delta\text{Welfare} - (\pi_B - FC) \quad (7)$$

$$= \underbrace{(-\Delta\text{TravelCosts})}_{>0} + \underbrace{(MC_A - p)/2}_{<0} + \underbrace{(\Delta\text{Health})}_{>0 \text{ or } <0} \quad (8)$$

Hospital B's OB provision can generate three externalities: a positive access-to-care externality (giving mothers access to an additional option), a negative business stealing externality (reduction in Hospital A's variable profit), and an ambiguously-signed maternal and infant health externality (changes in mothers' welfare from health). If the net externality is positive, it is possible for the unregulated market to have too little OB provision. If the net externality is negative, it is possible for the unregulated market to have too much OB provision.

4 Model of Demand, Health, and Prices

In this section, I develop an empirical model of OB hospital demand, maternal and infant health, and commercial price-setting in the hospital OB care market. I discuss identification, estimation, and results. For now, hospitals' OB provision decisions are taken as given (in Section 6, I extend the model to include them).

Outcomes are determined in two stages. In Stage 1, demand-based quality, health-based quality, and marginal cost shocks are realized. Then, commercial prices are set by bargaining between commercial insurers and hospital systems. In Stage 2, preference and underlying health shocks are realized. Then, mothers choose OB hospitals to receive care from; simultaneously, maternal and infant health is determined.

I proceed through the stages in reverse order, outlining their structures, how their parameters are identified and estimated, and estimation results. I begin by discussing Stage 2, OB hospital demand and maternal and infant health. Then, I discuss Stage 1, commercial price-setting.

Mothers are indexed by $i \in I$, insurers are indexed by $m \in M$, and hospitals are indexed by $j \in J$. **Bold** characters represent vectors.

4.1 Demand and Health

In Stage 2, preference and underlying health shocks are realized. Then, mothers choose OB hospitals to receive care from; simultaneously, maternal and infant health is determined. Prior to this stage, hospitals' OB provision decisions have been made, demand-based and health-based quality shocks have been realized, and commercial prices have been set.

Each mother is covered by a commercial insurer or Medicaid. Mother i 's insurer, denoted $m(i)$, is taken as exogenous.⁴ Let J_m^{OB} denote the set of OB hospitals that are in-network for insurer m . All hospitals are in-network for Medicaid, and networks are taken as exogenous.⁵

Let r_i denote the resource use intensity for mother i 's birth, normalized to be 1 on average. Each in-network insurer-OB hospital pair has price p_{mj} . The payment for mother i 's birth is $r_i p_{m(i)j}$. In other words, p_{mj} is price per unit of intensity. Given the payment for mother i 's birth, she pays $\text{OOP}_i(r_i p_{m(i)j})$ out-of-pocket, where the function $\text{OOP}_i(\cdot)$ is determined by her deductible, copay, and coinsurance rate. r_i and $\text{OOP}_i(\cdot)$ are taken as exogenous. In Appendix E, I detail how r_i , p_{mj} , and $\text{OOP}_i(\cdot)$ are recovered from the data.

Finally, mothers' home locations are taken as exogenous.⁶ Let Dist_{ij} denote the distance between mother i 's home and hospital j .

Each mother maximizes her utility by choosing among her choice set: in-network OB hospitals plus an outside option (which encompasses possibilities such as out-of-network OB hospitals, non-OB hospital emergency rooms, birth centers, or home births). Mother i 's utility from choosing OB hospital j is

$$u_{ij} = -\alpha \text{OOP}_i(r_i p_{m(i)j}) - \beta_i (\text{Dist}_{ij} + \eta \text{Dist}_{ij}^2) + \gamma_i' \mathbf{y}_j + \xi_j \quad (9)$$

where $\mathbf{y}_j$ are hospitals' characteristics and ξ_j is an unobserved demand-based quality shock. The random coefficients β_i and γ_i are parameterized as

$$\beta_i = \exp(\delta_0^{\text{Dist}} + \boldsymbol{\delta}^{\text{Dist}'} \mathbf{x}_i + \nu_i^{\text{Dist}}) \quad (10)$$

$$\gamma_i^k = \delta_0^k + \boldsymbol{\delta}^{k'} \mathbf{x}_i + \nu_i^k \quad \text{for all hospital characteristics } k \quad (11)$$

where $\mathbf{x}_i$ are mothers' characteristics and ν_i are unobserved preference shocks.

Simultaneously, maternal and infant health is determined. Let Healthy_{ij} be an indicator for

⁴In Appendix C, I present descriptive evidence suggesting that mothers do not choose insurers for reasons related to OB care.

⁵Network formation happens at the hospital level and is unlikely to be driven by OB care. Among OB hospitals in this setting, the average share of revenue from OB care is 3.2%.

⁶In Appendix D, I present descriptive evidence suggesting that conditional on observed characteristics, mothers' home locations are unrelated to OB care.

whether mother i choosing OB hospital j experiences no maternal or infant adverse outcomes. Healthy_{ij} is

$$\text{Healthy}_{ij} = \mathbb{1}(\xi' \mathbf{x}_i + \kappa' \mathbf{y}_j + \mu' \mathbf{z}_{ij} + \lambda_j + \epsilon_i > 0) \quad (12)$$

where $\mathbf{x}_i$ are mothers' characteristics, $\mathbf{y}_j$ are hospitals' characteristics, $\mathbf{z}_{ij}$ are interacted mother-hospital characteristics, λ_j is an unobserved health-based quality shock, and ϵ_i is an unobserved underlying health shock.

I parameterize the joint distributions of unobserved shocks:

$$\begin{pmatrix} \mathbf{v}_i \\ \epsilon_i \end{pmatrix} \sim N(0, \Sigma_{\mathbf{v}\epsilon}) \quad (13)$$

$$\begin{pmatrix} \xi_j \\ \lambda_j \end{pmatrix} \sim N(0, \Sigma_{\xi\lambda}) \quad (14)$$

Choice Probabilities, Expected Utility, and Willingness to Pay Given this structure, the probability that mother i chooses OB hospital j is

$$s_{ij}(J_{m(i)}^{\text{OB}}, \mathbf{p}_{m(i)}) = \Pr \left(j = \underset{k \in J_{m(i)}^{\text{OB}} \cup \{\text{outside}\}}{\text{argmax}} u_{ik} \mid \xi \right) \quad (15)$$

and mother i 's expected utility is

$$\text{EU}_i(J_{m(i)}^{\text{OB}}, \mathbf{p}_{m(i)}) = \mathbb{E} \left[\max_{j \in J_{m(i)}^{\text{OB}} \cup \{\text{outside}\}} u_{ij} \mid \xi \right] \quad (16)$$

where uncertainty is over preference shocks $\mathbf{v}_i$.

Mother i 's willingness to pay for a set of in-network OB hospitals J' is

$$\text{WTP}_{iJ'} = \frac{1}{\alpha} [\text{EU}_i(J_{m(i)}^{\text{OB}}, \mathbf{p}_{m(i)}) - \text{EU}_i(J_{m(i)}^{\text{OB}} \setminus J', \mathbf{p}_{m(i)})] \quad (17)$$

where $\frac{1}{\alpha}$ converts expected utility to monetary units.

Discussion of Demand and Health Model I discuss two notable features of the model of OB hospital demand and maternal and infant health. First, I address selection, a common issue in health outcome models: Healthy_{ij} is only observed for ij pairs where mother i chooses OB hospital j . This poses a problem if a mother's unobserved preferences are correlated with her unobserved underlying health. For example, if mothers who are unobservably sicker (lower ϵ_i) are unobservably more likely to choose hospitals with characteristic k (higher $\mathbf{v}_i^k$), then

naive estimation of Equation 12 (e.g., probit regression) would underestimate κ^k : hospitals with characteristic k would appear lower-quality because they attract sicker mothers.

To account for selection, I leverage the demand model and explicitly model correlations between unobserved preferences and unobserved underlying health in $\Sigma_{\nu\epsilon}$. Then, if sicker mothers are more likely to choose hospitals with characteristic k , the correlation would be captured by $\text{Cov}(\nu_i^k, \epsilon_i)$ instead of biasing the estimate of κ^k .

Second, I model demand as pure characteristics, omitting from Equation 9 an unobserved ij -level (usually Gumbel-distributed) utility shock that often appears in discrete choice models. Instead, all mother-level unobserved preferences are contained in the preference shocks ν_i and are thus disciplined by hospitals' characteristics. I do so for two reasons.

One reason is that demand models with ij -level shocks, which imply the characteristic space expands without bound as the number of choices increases, tend to overstate the value of larger choice sets (Ackerberg & Rysman, 2005; Berry & Pakes, 2007). Because the value of larger choice sets is central to my research question, omitting an ij -level shock is a conservative measure. Another reason is that an ij -level shock opens up an additional channel of unobserved preferences, and allowing this channel to correlate with underlying health shocks ϵ_i is infeasible: correlations between ij -level shocks and ϵ_i introduce too many parameters (as many as the number of OB hospitals) to reliably identify given the rarity of adverse outcomes.⁷ A pure characteristics model makes it feasible to force all unobserved preferences into a channel that correlates with ϵ_i , capturing any possible selection.

The downside of a pure characteristics model is computational complexity, as an ij -level shock provides helpful smoothing properties. I use methods from Cuesta and Tebaldi (2025), who show that rich individual-level data provide similar smoothing properties that make computation tractable.

Identification of Demand and Health Parameters I normalize $\alpha = 1$ and $\text{Var}(\epsilon_i) = 1$. The parameters to identify are $\{\delta_0^{\text{Dist}}, \delta^{\text{Dist}}\}$, $\{\delta_0^k, \delta^k\}$ for all hospital characteristics $k, \eta, \zeta, \kappa, \mu$, and $\Sigma_{\nu\epsilon}$. I also identify $\Sigma_{\xi\lambda}$ and the realized values of $\{\xi_j, \lambda_j\}$ for all OB hospitals j .

For all mothers i and OB hospitals j , I observe $\text{OOP}_i(r_i p_{m(i)j})$, Dist_{ij} , $\mathbf{x}_i$, $\mathbf{y}_j$, and $\mathbf{z}_{ij}$. I also observe networks J_m^{OB} and choices $\mathbb{1}(i \text{ chooses } j)$. Finally, I observe maternal and infant health at chosen OB hospitals, or $\text{Healthy}_{ij(i)}$.

δ_0^{Dist} is identified by how choice probabilities vary with distance, and δ^{Dist} are identified by how that distance preference varies with mothers' characteristics. Similarly, for each hospital characteristic k , δ_0^k is identified by how choice probabilities vary with k , and δ^k are identified

⁷Geweke et al. (2003) estimate such a model using Bayesian inference methods in a setting with a higher adverse outcome rate than the setting of this paper.

by how that preference for k varies with mothers' characteristics. η is identified by the distance preference's curvature. $\text{Var}(\nu_i)$ are identified by choices not explained by common preferences. For example, suppose some mothers choose their nearest option even if better options exist slightly farther away, while other mothers travel far to choose their best options; this implies $\text{Var}(\nu_i^{\text{Dist}})$ is large.

ζ , κ , and μ are identified by how health varies with observed characteristics. For characteristic k , $\text{Cov}(\epsilon_i, \nu_i^k)$ is identified by how health varies with unobserved preference for k . To separately identify the health effect of k and selection through k , this unobserved preference must be inferred from exogenous choice shifters. In the model, the main choice shifter is home location. For example, consider two mothers living in different locations (with otherwise identical characteristics) who both choose hospital j and travel the same distance to reach it; if one of them lives near many other options while the other does not, they likely have different unobserved preferences. In addition to home locations, two more choice shifters exist for commercial mothers: out-of-pocket spending and networks.

ξ_j is identified by the common preference for hospital j not explained by observed characteristics, and $\text{Var}(\xi_j)$ is identified by the spread of those preferences. Similarly, λ_j is identified by the common health effect of hospital j not explained by observed characteristics, and $\text{Var}(\lambda_j)$ is identified by the spread of those health effects. $\text{Cov}(\xi_j, \lambda_j)$ is identified by the correlation between those preferences and health effects.

Estimation of Demand and Health Parameters For mothers' characteristics in Equations 10 and 11, I use a Medicaid indicator, an ARHOME indicator, mother's age, mother's BMI, a prior birth indicator, a prior caesarean section indicator, and logged home county population density. For hospitals' characteristics in Equation 9, I use an outside option indicator, a UAMS indicator, a WCWH indicator, a NICU indicator, logged non-OB beds, and an index constructed using reviews from Medicare patient surveys.

In Equation 12, maternal and infant health is defined as an indicator for no adverse outcomes. For mothers' characteristics, I use a constant, a Medicaid indicator, an ARHOME indicator, mother's age, mother's race, mother's BMI, a diabetes indicator, a hypertension indicator, a smoker indicator, a prior birth indicator, a prior caesarean section indicator, logged home county population density, and home county population shares by income bin. For hospitals' characteristics, I use an outside option indicator, a UAMS indicator, a NICU indicator, logged non-OB beds, and an index constructed using outcomes for Medicare patients (specifically, mortality and readmission rates for pneumonia, COPD, and heart failure). For interacted mother-hospital characteristics, I use a selected set of hospitals' characteristics (UAMS, NICU, logged non-OB beds, and the patient outcomes index) interacted with a selected set of mothers'

characteristics (age, BMI, prior birth, and prior caesarean section). I also use distance and distance interacted with those same mothers' characteristics.

I assume the outside option has the insurer-level average price, which determines its out-of-pocket spending. Other than out-of-pocket spending and the outside option indicator, the outside option's characteristics are set to 0. I assume $\delta^{\text{outside}} = \delta^{\text{WCWH}} = 0$ and $\nu_i^{\text{outside}} = \nu_i^{\text{WCWH}} = 0$ for all mothers i ; in other words, there is no observed or unobserved heterogeneity in preferences for either the outside option or WCWH. Furthermore, I assume $\text{Cov}(\nu_i)$ is diagonal; in other words, unobserved preferences for different hospital characteristics are independent of each other.

Because common preferences for the outside option, UAMS, and WCWH are entirely captured by the coefficients on their indicators, leaving no room for unobserved components, I normalize ξ_{outside} , ξ_{UAMS} , and ξ_{WCWH} to 0. For the same reason in health effects, I normalize λ_{outside} and λ_{UAMS} to 0.

I apply two steps of maximum likelihood estimation. In the first step, using observed choices, I estimate the demand parameters. I estimate hospital-specific fixed effects, which are each separated by generalized least squares into the component explained by observed characteristics (to obtain δ_0^k for all hospital characteristics k) and the unobserved component (to obtain σ_ξ and ξ_j for all OB hospitals j). In the second step, I take the demand estimates as given, including estimates of ξ_j adjusted with empirical Bayes shrinkage. Using observed choices and health outcomes, I estimate the health parameters, including λ_j as a random effect. Second-step standard errors are corrected for first-step imprecision by drawing demand parameters from their covariance. Table 3 shows the main results, which are related to distance and hospital size. In Appendix F, I present the complete set of results.

Mothers strongly prefer nearby hospitals: the median value that a mother places on an additional mile of distance is equivalent to \$226 more out-of-pocket spending.⁸ However, the effect of distance on maternal and infant health is small. An additional mile lowers the probability of experiencing no adverse outcomes by 0.02 percentage points. This suggests mothers value proximity for reasons unrelated to health.

Hospital size, measured by the number of non-OB beds, plays a role in demand. Mothers prefer larger hospitals: on average, mothers value a hospital with twice as many non-OB beds equivalently to \$3,500 less out-of-pocket spending. This preference is stronger for older mothers and mothers with higher BMIs (who may expect to have more complicated deliveries), and it is weaker for mothers with prior births without caesarean sections (who may expect to have less complicated deliveries). In contrast, I find no evidence of a systematic effect of hospital

⁸I highlight medians instead of averages because distance effects are skewed given the lognormal distribution of β_i . The average value of distance is equivalent to \$940 more out-of-pocket spending per mile.

Table 3: Main Demand and Health Results

	Median Δ Utility (\$)	Avg Δ Pr(Healthy) (%)
Distance (mi)	−225.8 (69.7)	−0.0213 (0.0092)
Δ Effect of Distance (mi) per Δx_i :		
Age (yr)	0.1 (1.8)	−0.0006 (0.0007)
BMI	1.8 (1.4)	0.0010 (0.0005)
Prior Births>0	−42.9 (25.3)	0.0100 (0.0098)
Prior C-sections>0	40.1 (24.3)	0.0001 (0.0138)
	Avg Δ Utility (\$)	Avg Δ Pr(Healthy) (%)
2×Non-OB Beds	3486.4 (1862.3)	−0.190 (0.400)
Δ Effect of 2×Non-OB Beds per Δx_i :		
Age (yr)	119.8 (29.2)	−0.001 (0.014)
BMI	57.3 (15.4)	0.023 (0.013)
Prior Births>0	−506.3 (148.3)	0.166 (0.178)
Prior C-sections>0	633.7 (374.8)	0.130 (0.287)
Observations	20087	

Sample: births with claims data. Utility effects of distance are median marginal effects evaluated at chosen distances. All other effects are average marginal effects. County-clustered SEs for estimates related to mother-level variation, model-based SEs for estimates related to only hospital-level variation. Additional results in Appendix F.

size on maternal and infant health. Just like distance, mothers value hospital size for reasons unrelated to health.

Finally, I quantify the extent to which mothers' choices are driven by health in a cross-hospital comparison. I define each hospital j 's demand-based quality as the average mother's increase in utility from choosing hospital j relative to the outside option, setting distance and out-of-pocket spending to zero. In other words, I measure how much mothers would value hospital j if it were in-network, free, and required no travel. Then, I define each hospital j 's health-based quality as the average mother's change in no-adverse-outcome rate relative to the outside option, setting distance to zero. In other words, I measure how much hospital j would improve maternal and infant health if it required no travel.

Figure 3 shows the relationship between demand-based and health-based quality. Even though they are positively related, the relationship is noisy, and the slope is not statistically significant. Health-based quality explains only 3% of variation in demand-based quality. This result is consistent with mothers not fully internalizing differences in maternal and infant health when choosing OB hospitals (e.g., due to lack of information about health-based quality). In social welfare calculations, I assume the value of maternal and infant health is not captured by mothers' demand-based welfare and include it as a separate component of social welfare.

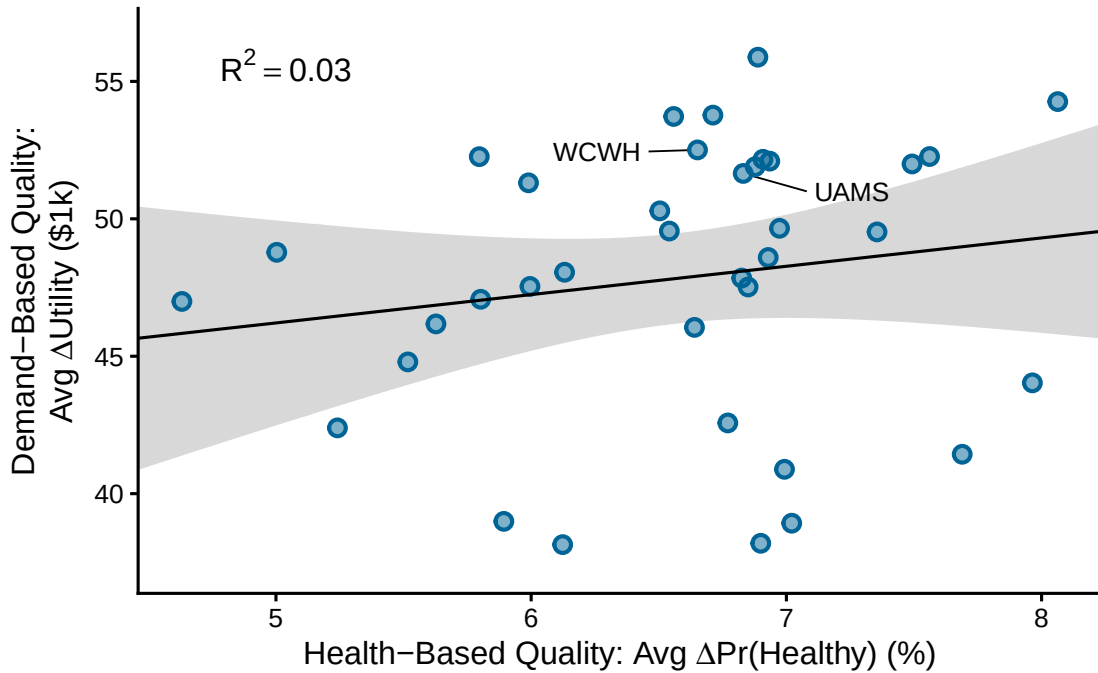


Figure 3: Relationship Between Demand-Based and Health-Based Quality

Sample: OB hospitals. Health-based quality is average change in no-adverse-outcome rate relative to outside option, ignoring distance. Demand-based quality is average change in utility relative to outside option, ignoring distance and OOP. Labeled hospitals are UAMS (teaching) and WCWH (women's). Sloped line from OLS regression. Shaded area is 95% CI, taking quality estimates as given.

4.2 Commercial Price-Setting

In Stage 1, health-based quality, demand-based quality, and commercial marginal cost shocks are realized. Then, commercial prices are set by bargaining between commercial insurers and hospital systems. Prior to this stage, hospitals' OB provision decisions have been made. Insurers and systems have expectations about OB hospital demand, where uncertainty is over unrealized preference shocks ν_i for all mothers i . Let I_m denote the set of mothers covered by insurer m and J_s^{OB} denote the set of OB hospitals that belong to system s .

Each insurer is a perfect agent for its enrollees: it values its enrollees' expected utility but faces costs from expected payments net of out-of-pocket spending. Specifically, insurer m 's value (as a function of its in-network OB hospitals J_m^{OB} and prices $\mathbf{p}_m$) is

$$V_m(J_m^{\text{OB}}, \mathbf{p}_m) = \frac{1}{\alpha} \sum_{i \in I_m} \text{EU}_i(J_m^{\text{OB}}, \mathbf{p}_m) - \sum_{i \in I_m} \sum_{j \in J_m^{\text{OB}}} [s_{ij}(J_m^{\text{OB}}, \mathbf{p}_m)(r_i p_{mj} - \text{OOP}_i(r_i p_{mj}))] \quad (18)$$

where $\frac{1}{\alpha}$ converts expected utility to monetary units.

I assume markups (prices net of marginal costs) are constant within an insurer-system pair. If system s is in-network for insurer m , then they engage in Nash-in-Nash bargaining over $\text{Markup}_{ms} = p_{mj} - \text{MC}_{mj}$ for all $j \in J_s^{\text{OB}}$. Given all other prices, their markup maximizes the exponentiated product of insurer m 's gain in value and system s 's gain in expected variable profit from network inclusion. Specifically, Markup_{ms} maximizes

$$\text{NB}_{ms}(\text{Markup}_{ms} \mid \mathbf{p}_{m,-s}) = [V_m(J_m^{\text{OB}}, \mathbf{p}_m) - V_m(J_m^{\text{OB}} \setminus J_s^{\text{OB}}, \mathbf{p}_{m,-s})]^b \left[\sum_{i \in I_m} \sum_{j \in J_s^{\text{OB}}} [s_{ij}(J_m^{\text{OB}}, \mathbf{p}_m) r_i] \text{Markup}_{ms} \right]^{1-b} \quad (19)$$

where $b \in [0, 1]$ is the bargaining weight of insurers relative to hospital systems and MC_{mj} is marginal cost per unit of intensity.

Given this structure, each commercial price p_{mj} satisfies

$$p_{mj} = \text{MC}_{mj} - \left[\frac{\partial Q_{ms}}{\partial \text{Markup}_{ms}} - \left(\frac{b}{1-b} \right) \left(\frac{\partial V_m(J_m^{\text{OB}}, \mathbf{p}_m) / \partial \text{Markup}_{ms}}{V_m(J_m^{\text{OB}}, \mathbf{p}_m) - V_m(J_m^{\text{OB}} \setminus J_s^{\text{OB}}, \mathbf{p}_{m,-s})} \right) \right]^{-1} Q_{ms} \quad (20)$$

where $Q_{ms} = \sum_{i \in I_m} \sum_{j \in J_s^{\text{OB}}} [s_{ij}(J_m^{\text{OB}}, \mathbf{p}_m) r_i]$.

I further parameterize marginal cost MC_{mj} as

$$\text{MC}_{mj} = \boldsymbol{\psi}' \mathbf{w}_{mj} + e_{mj} \quad (21)$$

where $\mathbf{w}_{mj}$ are insurer-hospital characteristics and e_{mj} is an unobserved marginal cost shock. I assume $e_{mj} \sim N(0, \sigma_e^2)$. I also assume e_{mj} is independent of hospital j 's demand-based and health-based quality shocks.

Discussion of Commercial Price-Setting Model I discuss two topics related to the model of commercial price-setting. First, marginal cost may include not only direct accounting costs (e.g., labor, equipment, or medical supplies), but also indirect costs and benefits internalized

by hospital systems. For example, marginal costs may be lowered by public and non-profit hospitals' mission-guided incentives to provide care, or by OB patients consuming other services at the same hospital.

Second, I make simplifying assumptions about the model's bargaining framework. I assume insurers are perfect agents that value enrollees' utility one-for-one with payments, and I assume markups are constant within an insurer-system pair. In the literature on models of hospital markets, insurers are often modeled as imperfect agents with a relative utility weight, and multi-hospital systems often flexibly divide markups among their hospitals (Gowrisankaran et al., 2015). However, these two features, an insurer's utility weight and a system's division of markups, would be identified solely by variation in demand elasticity with respect to price. In this setting, out-of-pocket spending is zero for many mothers and near zero for the rest, so there is not enough variation for reliable identification.

Furthermore, I assume a common relative bargaining weight b for all insurer-system pairs. I do so because cross-insurer and cross-hospital variation in bargaining weights would be difficult to identify alongside variation in marginal costs. I choose to focus on marginal costs, which are more closely related to my research question than bargaining weights are.

Identification of Commercial Price-Setting Parameters The parameters to identify are b , ψ , σ_e , and the realized values of e_{mj} for all in-network commercial insurer-OB hospital pairs m, j . For all mothers i and OB hospitals j , I observe r_i , $p_{m(i)j}$, $OOP_i(\cdot)$, and $\mathbf{w}_{m(i)j}$. In addition, $s_{ij}(\cdot)$ and $EU_i(\cdot)$ are identified from the OB hospital demand stage.

For candidate values of b and ψ , Equation 20 gives implied values of marginal costs MC_{mj} , and then Equation 21 gives implied values of marginal cost shocks e_{mj} . The true parameter values uniquely satisfy

$$E[e_{mj}(b, \psi) \tilde{\mathbf{w}}_{mj}] = 0 \quad (22)$$

where $\tilde{\mathbf{w}}_{mj}$ are instruments plausibly assumed to be related to price but independent of e_{mj} . The moments in Equation 22 for all in-network commercial insurer-OB hospital pairs identify b , ψ , and e_{mj} for all pairs. The spread of e_{mj} identifies σ_e .

$\tilde{\mathbf{w}}_{mj}$ consist of $\mathbf{w}_{mj}$ along with an additional instrument: for insurer m and hospital j , the average willingness to pay among insurer m 's enrollees for hospital j 's system if all prices were equal to the average observed price.

The following explanations provide intuition for how the parameters are identified. First, ψ are identified by how price varies with $\mathbf{w}_{mj}$. Second, b is identified by how price varies with the willingness to pay instrument. If b is low, then hospital systems have more bargaining power; more desirable systems can command higher prices, so willingness to pay is positively related to price. If b is high, then insurers have more bargaining power; prices are driven down to

marginal costs, so willingness to pay is less closely related to price. Third, marginal cost shocks e_{mj} are identified by the components of price not explained by observed characteristics, and σ_e is identified by their spread.

Estimation of Commercial Price-Setting Parameters For insurer-hospital characteristics in Equation 21, I use a constant, a UAMS indicator, a NICU indicator, logged non-OB beds, a for-profit indicator, the CMS Wage Index (which measures regional hospital worker wages), and an indicator for ARHOME. When computing gains from network inclusion, I assume the price of the outside option is the insurer-level average price.

Using empirical analogs of the moments in Equation 22, I apply generalized method of moments. Standard errors are clustered by insurer and system, and they take demand estimates as given. Table 4 shows the results.

Table 4: Commercial Price-Setting Results

	Marginal Cost (\$1k)
Constant	5.412 (1.532)
UAMS	0.909 (0.432)
NICU	0.115 (0.599)
2×Non-OB Beds	−0.260 (0.456)
For-Profit	−1.190 (1.516)
Wage Index (1 SD)	1.800 (1.144)
ARHOME	−1.356 (0.386)
Insurers' Relative Bargaining Weight	0.613 (0.170)
SD of Marginal Cost Shock	1.996 (0.333)
Observations	1376

Sample: in-network commercial insurer-OB hospital pairs. Pairs where insurer's gain from network inclusion is not positive (8 pairs) and pairs with no births with claims data (10 pairs) are dropped. Marginal cost per unit of intensity. SEs clustered by insurer and system, taking demand estimates as given.

Births at UAMS, the only hospital with an obstetrics and gynecology residency program, cost \$900 more per unit of intensity than births at otherwise identical hospitals. Medicaid-funded

ARHOME births cost \$1,400 less per unit of intensity than other commercial births. Insurers' relative bargaining weight is 0.6, suggesting that insurers capture somewhat more than half of the surplus from network inclusion, although I cannot reject an even split. The estimates imply hospitals earn large markups. Among non-outside-option commercial births, the average marginal cost is \$4,856, and the average markup is \$3,637.

Medicaid Prices and Marginal Costs Counterfactual simulations require Medicaid prices and marginal costs. Medicaid prices for OB hospitals are observed and taken as exogenous. For a hospital that does not operate an OB unit, I assume its Medicaid price if it did operate one would be the average observed Medicaid price.

To infer Medicaid marginal costs, I assume hospital j 's Medicaid marginal cost is

$$MC_{\text{Medicaid},j} = \psi' \mathbf{w}_{\text{ARHOME},j} - \psi^{\text{Medicaid}} \quad (23)$$

In other words, a Medicaid birth's marginal cost at hospital j is the same as that of the average ARHOME birth at hospital j , minus a Medicaid-specific cost reduction. ARHOME is a natural benchmark because ARHOME plans are Medicaid-funded and cover residents with incomes close to the Medicaid threshold. A plausible explanation for the cost reduction is supplemental payments, annual lump-sum payments that hospitals receive for treating Medicaid patients. In Arkansas, these payments reimburse hospitals based on how much more revenue they would have earned with Medicare, instead of Medicaid, prices (ACHI, 2023). Thus, a hospital internalizes a lower marginal cost for a Medicaid birth than for an otherwise identical non-Medicaid birth.

Using supplemental payments data, I compute supplemental payments per Medicaid inpatient day among all Arkansas hospitals in 2022. Then, I multiply it by the average number of inpatient days per birth in the sample. I define the result as ψ^{Medicaid} .

The resulting estimate implies a Medicaid birth's marginal cost is \$2,238 lower than that of the average ARHOME birth at the same hospital. Implied Medicaid markups are large: among non-outside-option Medicaid births, the average marginal cost is \$1,187, and the average markup is \$1,819.

Outside Option Prices and Marginal Costs Counterfactual simulations also require prices and marginal costs of the outside option. For insurer m , I assume the outside option's price and marginal cost are equal to the average price and marginal cost across insurer m 's in-network insurer-OB hospital pairs. The assumed values stay fixed in all counterfactuals.

5 Effects of OB Provision

In this section, I apply the estimated model to answer the following questions. How does OB provision affect access to care, the variable costs of delivering OB care, and maternal and infant health? How does the social value of these effects compare with the private surplus that OB units generate? What is the net externality of OB provision? I focus on the 37 OB hospitals and quantify the effects of preventing their OB units from closing.⁹

I conduct counterfactual simulations with the following procedure. As a baseline, I simulate equilibrium outcomes under the status quo configuration of OB provision, using realized values of demand-based quality, health-based quality, and marginal cost shocks. Then, for each OB hospital j , I simulate the same equilibrium outcomes if hospital j did not operate an OB unit, holding fixed the OB provision of all other hospitals. The difference between those outcomes and baseline outcomes is the effect of hospital j 's OB provision.

First, I measure how each hospital's OB provision affects its private surplus, i.e., the expected variable profit that it generates for the system that owns it. Then, I measure the variable social surplus that it generates through the access-to-care, variable cost, and maternal and infant health effects. I compute the following changes: the change in mothers' demand-based welfare before out-of-pocket spending (i.e., expected utility, setting out-of-pocket spending to zero), the change in expected variable costs, and the change in expected births with no adverse outcomes.

I convert health effects to monetary units using the social cost of a preterm birth estimated by Waitzman et al. (2021). In doing so, I assume the marginal counterfactual-induced adverse-outcome birth is as costly as a preterm birth.¹⁰ Inflation-adjusting to 2022 dollars, the marginal social cost of an adverse-outcome birth is \$72,845.

Figure 4 compares an OB hospital's effect on private surplus with each of its three effects on social surplus. The horizontal axis describes an OB hospital's effect on its system's variable profit. The first plot compares it with the access-to-care effect, the second plot compares it with the variable cost effect, and the third plot compares it with the maternal and infant health effect. Each measure of social surplus is divided by the number of Arkansas births.

On the one hand, access-to-care effects are positive and large (on average, \$159 per Arkansas birth). They are larger for OB hospitals that attract more births and earn higher variable profits. These effects reflect the high value that mothers place on access to more options and shorter

⁹OB unit closures are frequent and OB unit openings are not. In Appendix J, I show the history of OB provision in this setting.

¹⁰An alternative approach is to use the social cost of the average adverse-outcome birth, computed by multiplying the value of a statistical life by the infant death share of adverse-outcome births. However, this would overstate health effects because the marginal adverse-outcome birth is less serious than the average adverse-outcome birth. The difference is likely large because the worst outcome, infant death, is far costlier than any other adverse outcome and sharply skews the cost distribution.

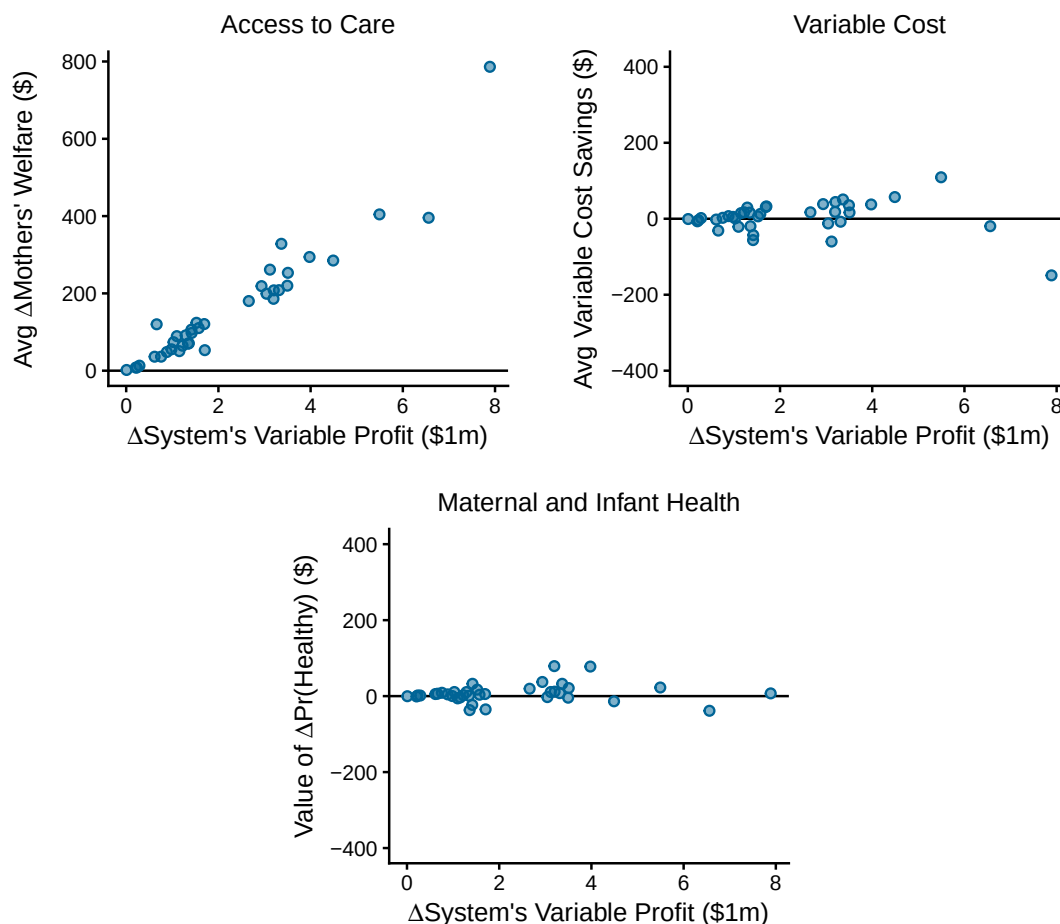


Figure 4: Access-to-Care, Variable Cost, and Maternal and Infant Health Effects

Sample: OB hospitals. Horizontal axis is change in system's expected variable profit caused by hospital's OB provision. Vertical axes are components of variable social surplus, each divided by Arkansas births: change in mothers' demand-based welfare before OOP, expected variable cost savings, and value of change in expected no-adverse-outcome births caused by hospital's OB provision.

travel distances. On the other hand, variable cost effects (on average, \$5 per Arkansas birth) and maternal and infant health effects (on average, \$7 per Arkansas birth) are mixed and small in magnitude. These effects are driven by the redirection of mothers between high-cost and low-cost, or high-quality and low-quality, hospitals. Some hospitals reduce variable costs or improve health by operating OB units, while others increase variable costs or worsen health by doing so. Given the sizes of these effects, access to care dominates: it accounts for 93% of the average sum of all three components.

The sum of all three components, aggregated over Arkansas births, is the total effect on variable social surplus. Figure 5 shows the relationship between an OB hospital's total effect on social surplus and its effect on private surplus. The difference between them is exactly the net externality of OB provision. In the figure, net externalities are depicted as distances between

an OB hospital's effect on variable social surplus, plotted on the vertical axis, and its effect on variable profit, denoted by the 45° line.

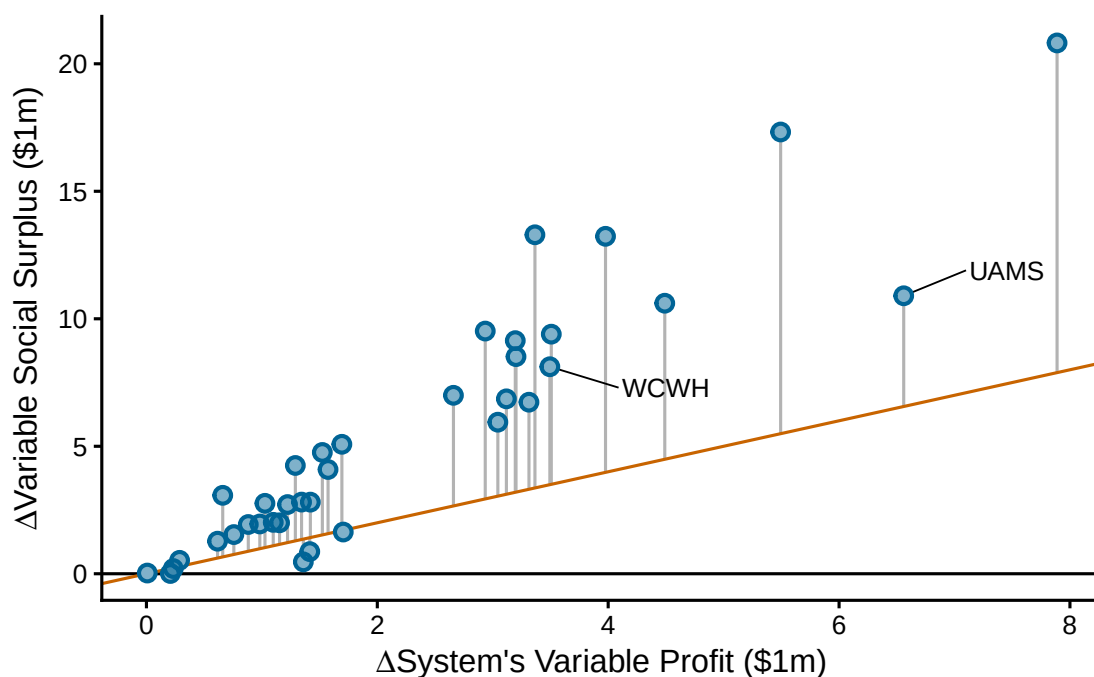


Figure 5: Net Externality of OB Provision

Sample: OB hospitals. Horizontal axis is change in system's expected variable profit caused by hospital's OB provision. Vertical axis is variable social surplus, summing the following: change in mothers' demand-based welfare before OOP, expected variable cost savings, and value of change in expected no-adverse-outcome births caused by hospital's OB provision. Sloped line is 45° line, and vertical lines denote net externalities. Labeled hospitals are UAMS (teaching) and WCWH (women's).

Almost all OB hospitals have positive externalities, and hospitals that attract more births have larger externalities. As suggested by the comparison in Figure 4, these externalities are largely driven by the access-to-care effect. On average, an OB hospital earns \$2.24 million in variable profit for its system and generates an additional externality of \$3.28 million (or \$102 per Arkansas birth). These large externalities suggest it is possible for an unregulated market to have too little OB provision relative to the welfare-maximizing outcome.

Finally, I discuss additional results, with associated figures being shown in Appendix G. First, Figure G1 shows heterogeneity in access-to-care and maternal and infant health effects. OB provision benefits rural mothers more than urban mothers. The average OB hospital improves mothers' demand-based welfare by \$214 per rural mother and \$128 per urban mother, and the average OB hospital improves no-adverse-outcome rates by 0.02 percentage points for rural mothers and 0.01 for urban mothers. Between Medicaid and commercial mothers, differences in effects are small.

Second, Figure G2 shows the distance effects of OB provision. Most OB hospitals reduce

travel distances, with a few exceptions that attract mothers from far away due to high demand-based quality. The average reduction is 0.10 miles per Arkansas birth. Distance effects are larger on rural mothers than urban mothers: average reductions are 0.18 miles per rural mother and 0.06 miles per urban mother.

Third, Figure G3 shows the role of distance in the access-to-care and maternal and infant health effects. For access to care, distance accounts for a substantial share of total effects, except for a few hospitals with high demand-based quality that increase travel distances but still provide value. For health, distance effects are near zero while total effects are larger in both positive and negative directions. Health effects are driven by variation in health-based quality, not distance.

Fourth, Figure G4 compares the net externalities of rural and urban OB hospitals. They are broadly similar: an average externality of \$3.1 million for rural hospitals and \$3.5 million for urban hospitals.

6 Model of OB Provision Decisions

In this section, I extend the empirical model of the hospital OB care market developed in Section 4 to include hospitals' OB provision decisions. Then, I use the model to identify and estimate the fixed cost of OB provision.

I add a Stage 0 that occurs before the two stages discussed in Section 4. In Stage 0, fixed cost shocks are realized. Then, hospital systems make OB provision decisions. Hospital systems have expectations about OB hospital demand, marginal costs, and commercial prices, where uncertainty is over unrealized preference shocks ν_i for all mothers i , demand-based quality shocks ξ , and marginal cost shocks $\mathbf{e}$.

The set of hospitals is taken as exogenous.¹¹ For hospital j , let M_j denote the set of insurers, including Medicaid, that have it in its network. Let $\text{Prov}_j \in \{0, 1\}$ denote hospital j 's OB provision decision. Given all hospitals' OB provision decisions, system s 's expected variable profit is

$$\pi_s(\mathbf{Prov}) = \mathbb{E} \left[\sum_{j \in J_s^{\text{OB}}} \sum_{m \in M_j} \sum_{i \in I_m} [s_{ij}(J_m^{\text{OB}}, \mathbf{p}_m) r_i(p_{mj} - \text{MC}_{mj})] \right] \quad (24)$$

For hospital j , given all other hospitals' decisions, let $\Delta_j \pi_{s(j)}(\mathbf{Prov}_{-j})$ denote the change in its system's expected variable profit if it switches from non-provision to provision. I assume systems play a pure-strategy subgame perfect equilibrium. This rules out profitable single-

¹¹Hospital openings and closures are rarer than OB unit closures, and they are unlikely to be driven by OB care. Among OB hospitals in this setting, the average share of revenue from OB care is 3.2%.

hospital deviations: given all other hospitals' decisions, hospital j operates an OB unit if the effect of its OB provision on its system's expected total profit is positive, and it does not if the effect is negative. In other words,

$$\text{Prov}_j = \begin{cases} 1 & \text{if } \Delta_j \pi_{s(j)}(\mathbf{Prov}_{-j}) > FC_j \\ 0 & \text{if } \Delta_j \pi_{s(j)}(\mathbf{Prov}_{-j}) < FC_j \end{cases} \quad (25)$$

where FC_j is fixed cost.

I further parameterize fixed cost as

$$FC_j = \overline{FC} + u_j \quad (26)$$

where $\overline{FC}$ is expected fixed cost and u_j is an unobserved fixed cost shock with $E[u_j] = 0$.

Discussion of OB Provision Model I highlight four notes about the model of OB provision. First, the OB provision decision is modeled as static, so it does not depend on past or future outcomes. In Appendix H, I discuss reasonable assumptions under which a static model captures all relevant features of OB provision.

Second, I follow much of the literature on fixed cost estimation in assuming demand-based quality and marginal cost shocks are not realized until after firms decide whether to participate in the market. Thus, participants and non-participants have the same distribution of those shocks.¹² To support this assumption, I include the most salient hospital characteristics that could shift demand or marginal costs in the utility and marginal cost equations (Equations 9 and 21), so it is plausible that any remaining variation is unanticipated before OB provision decisions are made.

Third, Equation 25 only rules out profitable single-hospital deviations. An equilibrium also rules out profitable multi-hospital deviations, such as a system switching out one OB hospital for another. Empirically, I find that ruling out multi-hospital deviations does not provide any additional information about fixed costs in this setting.

Fourth, just like marginal cost, fixed cost may reflect not just direct accounting costs (e.g., facilities, utilities, liability insurance), but also indirect costs or benefits from operating an OB unit for a year, such as reputational benefits.

¹²An alternative approach is to relax this assumption and estimate all stages of the model jointly, explicitly estimating the differences in those distributions (Ciliberto et al., 2021). In this paper, this is infeasible given the complexity of the OB hospital demand and commercial price-setting stages.

Identification of Expected Fixed Cost The parameter to identify is $\overline{FC}$. For all mothers i and hospitals j , I observe r_i and Prov_j . Also, the distributions of $s_{ij}(\cdot)$, $\text{MC}_{m(i)j}$, and $p_{m(i)j}$ are identified from the OB hospital demand and commercial price-setting stages.

Following Eizenberg (2014), I assume the support of fixed cost is in the support of the effect each hospital's OB provision has on its system's expected variable profit, given the observed OB provision of all other hospitals. In other words,

$$\text{FC}_j \in \left[\inf_{j' \in J} \Delta_{j'} \pi_{s(j')}(\mathbf{Prov}_{-j'}), \sup_{j' \in J} \Delta_{j'} \pi_{s(j')}(\mathbf{Prov}_{-j'}) \right] \quad \text{for all hospitals } j \quad (27)$$

This assumption is equivalent to the following statement: given the observed OB provision of all other hospitals, the most profitable hospital would operate an OB unit for any possible fixed cost shock, and the least profitable hospital would never operate an OB unit for any possible fixed cost shock.

This assumption, along with Equation 25, implies each FC_j has lower and upper bounds

$$L_j = \begin{cases} \inf_{j' \in J} \Delta_{j'} \pi_{s(j')}(\mathbf{Prov}_{-j'}) & \text{if } \text{Prov}_j = 1 \\ \Delta_j \pi_{s(j)}(\mathbf{Prov}_{-j}) & \text{if } \text{Prov}_j = 0 \end{cases} \quad (28)$$

$$U_j = \begin{cases} \Delta_j \pi_{s(j)}(\mathbf{Prov}_{-j}) & \text{if } \text{Prov}_j = 1 \\ \sup_{j' \in J} \Delta_{j'} \pi_{s(j')}(\mathbf{Prov}_{-j'}) & \text{if } \text{Prov}_j = 0 \end{cases} \quad (29)$$

Let $\tilde{\mathbf{y}}_j$ be instruments plausibly assumed to be related to OB provision but independent of fixed cost shock u_j . Then, Equation 26 implies

$$\overline{FC} \geq E[L_j \mid \tilde{\mathbf{y}}_j] \quad (30)$$

$$\overline{FC} \leq E[U_j \mid \tilde{\mathbf{y}}_j] \quad (31)$$

The identified set of $\overline{FC}$ is the set of values that satisfy the moment inequalities in Equations 30 and 31 for all values of $\tilde{\mathbf{y}}_j$. I include one instrument in $\tilde{\mathbf{y}}_j$: expected total intensity-weighted quantity if all hospitals operated OB units and all prices were equal to the average observed price, discretized into equal-sized bins.

For intuition, $\overline{FC}$ is identified by the relationship between OB provision and profitability: if expected fixed cost is higher, then hospitals need to earn higher variable profits to justify operating OB units. Conditioning on the instrument sharpens the identified set by forcing the estimate to explain differences in OB provision between attractive and unattractive hospitals under plausible restrictions on fixed cost shocks.

Estimation of Expected Fixed Cost I compute empirical analogs of the worst-case fixed cost bounds in Equation 27. I correct for the upward bias of a sample minimum and downward bias of a sample maximum following Hall and Park (2002). The corrected bounds are $-\$0.1$ million and $\$16.2$ million.

Using empirical analogs of the moment inequalities in Equations 30 and 31, I estimate expected fixed cost. An uncorrected estimated set would be the range between the highest average lower bound and the lowest average upper bound over all instrument bins. However, due to estimation error, the highest average is biased upward, and the lowest average is biased downward. I apply the estimation and inference procedure of Chernozhukov et al. (2013), which corrects each average according to its imprecision. Confidence intervals are clustered by system, and they take demand and commercial price-setting estimates as given.

Table 5 shows the results. The main specification discretizes the quantity instrument into 10 equal-sized bins. As a robustness check, I show how results vary for different numbers of bins. The main specification implies expected fixed cost is between $\$0.56$ million and $\$2.23$ million. This result does rely on the instrument, but it is not very sensitive to reasonable changes in the number of instrument bins.

Table 5: OB Provision Results

Instrument Bins	Expected Fixed Cost (\$1m)	
	Estimated Set	95% CI
1 (No Instrument)	[0.28, 9.01]	[0.17, 10.71]
8	[0.53, 3.56]	[0.24, 4.30]
9	[0.39, 3.38]	[0.04, 4.09]
10 (Main)	[0.56, 2.23]	[0.14, 2.65]
11	[0.42, 2.81]	[−0.09, 3.29]
12	[0.46, 1.77]	[−0.01, 2.13]
Observations	71	

Sample: all hospitals. Main specification is 10 bins. Bias-corrected estimates and CIs following Chernozhukov et al. (2013). System-clustered CIs, taking demand and commercial price-setting estimates as given.

As an out-of-sample check of model fit, I examine actual OB unit closures two years before and after the sample period of 2022. Two hospitals closed their OB units before, and four hospitals closed their OB units after.¹³ In Appendix I, I show how these hospitals compare with the set of all hospitals in terms of model-predicted profitability.

The closing hospitals all rank among the lower half. In particular, the hospitals that closed their OB units after 2022 are almost exactly the ones predicted by the model to be most at risk of

¹³Data on actual OB unit closures come from UAMS's Arkansas Maternal Health Scorecard.

OB unit closure: they are the least, third-least, fifth-least, and sixth-least profitable OB hospitals. This check is notable because actual OB unit closures are not used in estimation.

7 Evaluation of the Healthy Moms, Healthy Babies Act

In this section, I apply the extended model to evaluate Arkansas’s 2025 Healthy Moms, Healthy Babies Act. In particular, I evaluate the policy’s 70% increase in the prices paid by Medicaid for OB care. The motivation for this increase is to prevent OB unit closures, which are especially salient in Arkansas. In Appendix J, I show the history of OB provision in this setting: 24 hospitals closed their OB units between 1995 and 2024.

OB unit closures are likely driven by shrinking variable profits. As realistic baselines, I simulate 10%, 20%, and 30% declines in variable profits. I implement these declines by raising marginal costs enough to uniformly decrease all markups. Higher marginal costs align with one of the most commonly reported reasons for hospitals to close their OB units, the cost of retaining workers (Busse et al., 2025). Then, I simulate each of these declines along with uniform 10%, 35%, and 70% increases in all Medicaid prices. These represent incremental versions of the Healthy Moms, Healthy Babies Act.

In all counterfactuals, I focus on OB unit closures and not openings. I allow the 37 OB hospitals to change their OB provision decisions, but I assume the 34 non-OB hospitals never operate OB units. This is consistent with Arkansas’s historical trend of frequent OB unit closures without any openings, and it can be explained by OB provision having high one-time sunk costs, which I do not estimate.

I make the following assumptions about unobserved shocks. Hospital systems make OB provision decisions taking expectations over unrealized demand-based quality and marginal cost shocks. However, when computing outcomes, I assume demand-based quality, health-based quality, marginal cost, and fixed cost shocks always take on the same realized values as the status quo. The first three shocks are estimated; fixed cost shocks are not, so I assume a uniform prior distribution and take draws from the posterior distribution induced by observed OB provision.

Specifically, I use the following procedure. First, I choose a value of expected fixed cost $\overline{FC}$ from the estimated set. Then, I compute the set of all possible fixed cost shock vectors $\mathbf{u}$ that satisfy two conditions: observed OB provision is rationalized (i.e., $FC_j \in [L_j, U_j]$ for all hospitals j), and instrument exogeneity is empirically satisfied (i.e., $\text{Avg}(u_j \mid \tilde{\mathbf{y}}_j) = 0$ for all values of $\tilde{\mathbf{y}}_j$). I take draws of $\mathbf{u}$ from a uniform distribution over this set. For each draw, I solve for an equilibrium configuration of OB provision and simulate resulting outcomes. I take an average of those outcomes over draws of $\mathbf{u}$: this is the predicted outcome. Finally, I repeat this

procedure for all values of $\overline{FC}$ in the estimated set, and report the range of predicted outcomes.

In the step that requires solving for an equilibrium configuration of OB provision, I use a sequential algorithm. I start from the status quo configuration, and I compute each system's total-profit-maximizing configuration of its own hospitals, given the OB provision of all other hospitals. Then, I allow the system with the highest gain from deviating to change its configuration. I repeat this process, allowing one system to change its configuration at a time, until there are no more profitable deviations.¹⁴

One outcome of interest is total social welfare: mothers' demand-based welfare before out-of-pocket spending, plus the value of expected births with no adverse outcomes, minus expected variable cost, minus fixed cost. I also compute the marginal value of public funds, or the increase in private welfare per dollar of increased Medicaid spending (Hendren & Sprung-Keyser, 2020). Just like in Section 5, I convert health effects to monetary units by assuming the marginal social cost of an adverse-outcome birth is \$72,845, based on the estimated social cost of a preterm birth from Waitzman et al. (2021). Table 6 shows the results.

Table 6: Evaluation of the Healthy Moms, Healthy Babies Act

	OB Hospitals	Δ Social Welfare (\$1m)	MVPF
10% Variable Profit Decline	[31.4, 35.8]	—	—
+ Medicaid Price +10%	[32.4, 36.3]	[+1.0, +7.5]	[1.24, 2.83]
+ Medicaid Price +35%	[35.0, 37.0]	[+2.5, +15.4]	[1.17, 2.06]
+ Medicaid Price +70%	37.0	[+2.6, +21.4]	[1.09, 1.73]
20% Variable Profit Decline	[29.2, 34.4]	—	—
+ Medicaid Price +10%	[30.3, 35.0]	[+1.2, +3.1]	[1.27, 1.75]
+ Medicaid Price +35%	[32.2, 36.2]	[+3.7, +10.3]	[1.25, 1.71]
+ Medicaid Price +70%	[34.8, 36.9]	[+5.5, +19.7]	[1.19, 1.67]
30% Variable Profit Decline	[27.0, 33.1]	—	—
+ Medicaid Price +10%	[27.8, 33.7]	[+1.5, +2.8]	[1.36, 1.66]
+ Medicaid Price +35%	[30.2, 35.1]	[+4.5, +9.6]	[1.31, 1.65]
+ Medicaid Price +70%	[32.2, 36.4]	[+7.4, +17.7]	[1.25, 1.61]

Variable profits decline due to higher marginal costs. Actual policy is 70% Medicaid price increase. Reported values are predicted outcomes: averages over fixed cost shocks drawn from uniform posterior. I report the range of predicted outcomes across the estimated set of expected fixed cost. Welfare consists of mothers' demand-based welfare before OOP, plus value of expected no-adverse-outcome births, minus expected variable cost, minus fixed cost. MVPF = $1 + \Delta \text{Welfare} / \Delta \text{Medicaid Spending}$.

Without the policy, declining variable profits lead to OB unit closures. Of the 37 OB hospitals,

¹⁴Two issues may arise with this approach. First, the algorithm may not converge, possibly due to equilibrium nonexistence. Empirically, I find convergence in 99.2% of simulations. Second, the algorithm selects one equilibrium when multiple equilibria may exist. I find that alternative algorithms do often select different equilibrium configurations, but reported results, such as those in Table 6, do not change much.

31 to 36 remain after a 10% decline in variable profits, 29 to 34 remain after a 20% decline, and 27 to 33 remain after a 30% decline.

The policy prevents most of these closures. With a 70% increase in Medicaid prices, all 37 OB hospitals remain after a 10% decline, 35 to 37 remain after a 20% decline, and 32 to 36 remain after a 30% decline. The policy increases social welfare by \$3 to \$21 million after a 10% decline, \$6 to \$20 million after a 20% decline, and \$7 to \$18 million after a 30% decline. Across all declines and all values of expected fixed cost in the estimated set, the marginal value of public funds ranges from 1.1 to 1.7. In other words, each dollar of spending is a dollar transferred to hospitals plus \$0.1 to \$0.7 of additional social welfare.

Smaller versions of the policy have higher marginal values of public funds. A 35% increase in Medicaid prices has a marginal value of public funds of 1.2 to 2.1, and a 10% increase has 1.2 to 2.8. The first dollars of the increase go the furthest because they keep the most marginal OB hospitals open; further dollars still raise welfare, but they increasingly go to OB units that would have remained open anyway. This is suggestive of a potential refinement to the policy: subsidies directed at the OB hospitals most at risk of OB unit closures would be more cost-effective than an across-the-board price increase.

Finally, I discuss additional results, with associated tables being shown in Appendix K. First, Table K1 decomposes the policy's effects on the individual components of social welfare. Almost all of the gain is access to care: mothers' demand-based welfare increases by \$3.9 to \$28.4 million, about 90% of the total gain. Keeping OB units open gives mothers more options and reduces travel distances. The value of healthy births increases welfare by \$0.2 to \$2.1 million. Changes in variable costs affect welfare by -\$1.4 to \$2.3 million, and the fixed costs paid by the OB units kept open affect welfare by -\$11.4 to -\$1.7 million.

Second, Tables K2 and K3 show heterogeneity in the policy's access-to-care and maternal and infant health effects. The policy benefits rural mothers more than urban mothers. The average rural mother's welfare increases by \$180 to \$1,741, compared to \$86 to \$669 for the average urban mother, and the average rural mother's probability of a healthy birth increases by 0.02 to 0.20 percentage points, compared to 0.01 to 0.03 percentage points for the average urban mother. Between Medicaid and commercial mothers, the policy's effects are similar.

Third, Table K4 shows the policy's distance effects. The policy gives mothers nearer options, reducing the average mother's travel distance by 0.12 to 0.93 miles. The reduction is larger for rural mothers, 0.21 to 2.00 miles, than for urban mothers, 0.07 to 0.50 miles.

Fourth, Table K5 shows the role of distance in the policy's access-to-care and maternal and infant health effects. Distance accounts for about half of the access-to-care effect, and it accounts for about a quarter of the health effect. The rest of the effects can be attributed to the demand-based and health-based quality of the OB units kept open relative to their competitors.

Fifth, Table [K6](#) shows the policy's effects on rural OB hospitals, urban OB hospitals, and Medicaid spending. The effects on rural and urban OB hospitals are broadly similar. The policy increases Medicaid spending by \$29.1 to \$29.3 million, which is close to the policy's stated annual investment of \$26 million (Arkansas Governor, [2025](#)).

8 Conclusion

I study OB provision: hospitals' decisions to operate obstetric (OB) units and participate in the hospital OB care market. How does OB provision affect access to care, the costs of delivering care, and maternal and infant health in equilibrium? What is the net externality of OB provision? What are the consequences of a policy that increases Medicaid prices to encourage OB provision?

OB provision can have four welfare effects: a positive access-to-care effect, a negative fixed cost effect, an ambiguously-signed variable cost effect, and an ambiguously-signed maternal and infant health effect. The net externality of OB provision may be positive or negative, so an unregulated market may have too little or too much OB provision relative to the welfare-maximizing outcome.

To study the empirical setting of Arkansas in 2022, I combine births and medical claims data from the Arkansas All-Payer Claims Database. I estimate a model of the hospital OB care market in which hospital systems make OB provision decisions, commercial prices are set by bargaining between commercial insurers and hospital systems, mothers choose OB hospitals to receive care from, and maternal and infant health is determined. Then, I quantify the effects of OB provision and evaluate Arkansas's 2025 Healthy Moms, Healthy Babies Act.

I find that OB provision benefits mothers substantially through access to care. The average OB hospital's access-to-care effect is worth \$159 per Arkansas birth, driven by the high value that mothers place on access to more options and shorter travel distances, and these effects are larger for rural mothers than urban mothers. In contrast, variable cost effects and maternal and infant health effects are mixed and small. Overall, OB hospitals generate positive net externalities, on average worth \$102 per Arkansas birth.

This gap between social and private surplus suggests a role for policy. Under realistic declines in variable profits, many hospitals close their OB units. The Healthy Moms, Healthy Babies Act's 70% increase in Medicaid prices prevents most of these closures. The policy's marginal value of public funds is 1.1 to 1.7: each dollar of Medicaid spending is a dollar transferred to hospitals plus \$0.1 to \$0.7 in additional social welfare. Its benefits come largely through access to care, and they are larger for rural mothers than urban mothers.

Smaller Medicaid price increases generate more welfare per dollar because the first dollars of the increase keep the most marginal OB units open, while further dollars increasingly go

to OB hospitals that would have remained open anyway. This suggests that subsidies directed at the OB hospitals most at risk of OB unit closures would be more cost-effective than an across-the-board price increase.

There are several promising areas for future research related to this paper. First, I abstract away from capacity constraints and efficiency gains from higher patient volumes. These frictions create additional consequences of reallocating patients, potentially amplifying the effects of OB provision. Second, I assume hospitals cannot influence their own demand-based quality, health-based quality, or marginal costs. Future research, by modeling OB hospitals' decisions to invest in quality or efficiency, can evaluate policies that induce low-quality OB hospitals to become high-quality, or high-cost OB hospitals to become low-cost. Third, I do not consider spillover effects; for example, one hospital's OB provision may drive up labor costs for other OB hospitals. Not only would modeling such spillovers be informative about indirect effects within the hospital OB care market, but it would also illustrate how OB provision can have far-reaching implications in other parts of the economy.

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A Data Construction

I only include mothers who live in the setting: mothers who live outside of Arkansas or in metropolitan areas with population centers in other states (Crittenden, Miller, or Little River Counties) are dropped. For multiple births (e.g., twins or triplets), I drop births after the first-occurring one. Births that occur outside of in-network OB hospitals in the setting (e.g., out-of-network OB hospitals, OB hospitals in dropped counties, non-OB hospitals, birth centers, or home births) are assigned to the outside option.

I observe mothers' home locations at the county level. I measure the haversine distances between the centroids of census block groups and hospitals, construct each mother's probability of living in each census block group using her county and age, and impute mothers' expected distances to hospitals.

I assign each birth to claims data using the following procedure. First, I match on a person identifier (encrypted combination of mother's name, mother's date of birth, and mother's sex). Then, I filter matches to those where the claims' dates of insurance coverage and dates of service overlap with the date of birth. Then, I filter claims to those with diagnosis or procedure codes related to labor and delivery. Finally, I filter matches to those where the claims' National Provider Identifiers (NPIs) are linked to the birth's OB hospital, either by geographic proximity or by frequency of matching claims.

There are two reasons why births may not match to claims data. First, the claims data are not comprehensive. Mothers may be missing if they pay entirely out-of-pocket, are uninsured, or are insured by entities not required to report to the All-Payer Claims Database (e.g., commercial employer-sponsored self-insured plans, plans with fewer than 2,000 enrollees, Medicare, or Veterans Affairs). Second, the match between datasets may not be perfect due to ambiguity in mothers' names or other data collection errors.

I define an insurer as a combination of payer type (e.g., Medicare, ARHOME, or employer-sponsored), submitter (e.g., Blue Cross Blue Shield), group (e.g., employer), and plan. Among

payer-submitter-group-plan combinations with fewer than 10 births with claims data, I aggregate them into one insurer per payer-submitter-group. Among payer-submitter-group combinations with fewer than 10 births with claims data, I aggregate them into one insurer per payer-submitter. Among payer-submitter combinations with fewer than 10 births with claims data, I aggregate them into one insurer per payer. Finally, if any insurers still have fewer than 10 births with claims data, I drop them and their associated births from the sample.

I define hospital system s as in-network for insurer m if either of the following is true: there is at least one birth with claims data at system s from insurer m 's enrollees, or there are at least 10 non-birth-related claims (which I match to hospitals using NPIs and geographic proximity) at system s from insurer m 's enrollees. A system being in-network is equivalent to all of its hospitals being in-network. All hospitals are in-network for Medicaid.

Finally, for insurers with fewer than 2 in-network OB hospitals, I drop them and their associated births from the sample.

B Additional Summary Statistics

Table B1 shows birth summary statistics split by whether they match to claims data. Table B2 shows birth summary statistics by quarter of the year. Table B3 shows rates of different adverse outcomes. Table B4 shows hospital-level summary statistics.

C Descriptive Evidence About Insurer Choice

If mothers strategically choose insurers for reasons related to OB care (e.g., out-of-pocket spending or networks), then the empirical model's assumption that insurers are exogenous would be violated, and the model's estimates would be biased. I present two sets of descriptive evidence suggesting that this is not the case.

First, I examine how many insurers a mother may plausibly be able to choose from. Table C1 shows the number of insurers available to each mother under the following assumptions. First, a mother only chooses among insurers in her own payer type. In other words, a Medicaid mother can only choose Medicaid, an ARHOME mother only chooses among ARHOME plans, a state employee only chooses among state employee plans, and a mother whose plan is from the Affordable Care Act (ACA) marketplace only chooses among plans from the ACA marketplace. Second, a mother with employer-sponsored insurance has an additional restriction: she only chooses among plans in the same group (which typically corresponds to employer).

Medicaid mothers only have one choice, and mothers with employer-sponsored and state

Table B1: Birth Summary Statistics, by Match to Claims Data

	Commercial (Matched)	Commercial (Not Matched)	Medicaid (Matched)	Medicaid/Other (Not Matched)
Mean				
Distance (mi)	17.4	15.7	17.1	15.3
Any Adverse Outcome	0.03	0.02	0.03	0.03
Rural	0.34	0.25	0.39	0.31
Age (yr)	28.0	29.2	25.7	27.4
BMI	28.9	28.6	29.0	28.7
Prior Births>0	0.58	0.56	0.66	0.70
Prior C-sections>0	0.14	0.15	0.16	0.14
Prenatal Visits	10.8	11.2	9.3	9.2
Gestational Age (wk)	38.2	38.3	38.1	38.3
Planned C-section	0.22	0.21	0.22	0.20
Outside Option	0.01	0.00	0.01	0.08
UAMS (Teaching)	0.09	0.05	0.12	0.15
WCWH (Women's)	0.05	0.05	0.06	0.11
Public	0.10	0.06	0.15	0.18
Non-profit	0.72	0.76	0.65	0.52
For-profit	0.16	0.18	0.20	0.22
NICU	0.54	0.60	0.50	0.55
Non-OB Beds	296.1	308.3	250.6	230.0
Observations	9561	8442	10526	3778

Adverse outcomes: maternal (perineal laceration, blood transfusion, unplanned hysterectomy, uterine rupture) or infant (Apgar<4, death, injury, seizure).

employee insurance have few options (on average 1.3 and 7, respectively). Together, those mothers account for 64% of the sample. I cannot rule out the possibility that the remainder, ARHOME mothers and mothers who shop on the ACA marketplace, have many options (up to 23 and 19, respectively). However, there are likely additional restrictions, such as plans being region-locked or only offered at certain times of year, that I do not account for.

Second, I show that even if mothers have access to multiple insurer options, they do not strategically choose insurers for reasons related to OB care. For each birth in October 9 to December 31, I examine whether the mother was enrolled with the birth insurer before the start of pregnancy (280 days before birth). I also examine whether, at any point during pregnancy, the mother is enrolled with any insurer other than the birth insurer. Table C2 shows the results.

Most mothers (74% of commercial and 62% of Medicaid mothers) are enrolled with their birth insurers before the start of pregnancy, suggesting that those insurers are not chosen for reasons related to OB care.

Some mothers are enrolled with another insurer at some point during pregnancy (29% of commercial and 9% of Medicaid mothers), so they may have an opportunity to strategically switch from the other insurer. However, there is little evidence of such switching. Few mothers

Table B2: Birth Summary Statistics, by Quarter

	Jan–Mar	Apr–Jun	Jul–Sep	Oct–Dec
Mean				
Distance (mi)	16.7	16.4	16.9	16.4
Any Adverse Outcome	0.03	0.03	0.03	0.03
Payment (\$)	5614	5524	5477	5588
Out-of-pocket (\$)	552	509	470	443
Commercial	0.56	0.56	0.55	0.56
Medicaid	0.41	0.41	0.42	0.41
Other	0.04	0.02	0.03	0.03
Rural	0.33	0.32	0.34	0.31
Age (yr)	27.5	27.5	27.3	27.6
BMI	28.8	28.9	28.7	28.9
Prior Births>0	0.61	0.62	0.61	0.62
Prior C-sections>0	0.14	0.15	0.15	0.15
Prenatal Visits	10.1	10.3	10.2	10.4
Gestational Age (wk)	38.2	38.2	38.2	38.2
Planned C-section	0.21	0.22	0.22	0.21
Outside Option	0.02	0.02	0.02	0.02
UAMS (Teaching)	0.10	0.10	0.09	0.10
WCWH (Women's)	0.06	0.06	0.06	0.06
Public	0.12	0.12	0.11	0.12
Non-profit	0.68	0.68	0.69	0.68
For-profit	0.19	0.18	0.18	0.18
NICU	0.54	0.55	0.53	0.55
Non-OB Beds	275.3	277.0	271.6	283.2
Observations	7835	7776	8530	8166

Adverse outcomes: maternal (perineal laceration, blood transfusion, unplanned hysterectomy, uterine rupture) or infant (Apgar<4, death, injury, seizure).

(3% of commercial and 5% of Medicaid mothers) switch away from a less desirable insurer, i.e., an insurer with higher average out-of-pocket spending for OB care at the chosen OB hospital than the birth insurer, or an insurer where the chosen OB hospital is out-of-network. Most switching is between equal-out-of-pocket-spending insurers, and in fact 7% of commercial mothers switch away from lower-out-of-pocket-spending insurers. Thus, any insurer switching is likely unrelated to OB care (e.g., administrative churn).

Table B3: Adverse Outcome Rates

	Mean
Maternal	
Perineal Laceration	0.0065
Blood Transfusion	0.0055
Unplanned Hysterectomy	0.0005
Uterine Rupture	0.0003
Infant	
Apgar<4	0.0101
Death	0.0068
Injury	0.0005
Seizure	0.0002
Observations	32307

Apgar: overall infant health 5 min after birth, scored 0–10, <4 is critical.

D Descriptive Evidence About Home Locations

I present descriptive evidence suggesting that conditional on observed characteristics, mothers' home locations are unrelated to OB care, supporting the empirical model's assumption that home locations are exogenous. There are two types of possible violations that would be problematic. First, mothers who are unobservably more sensitive to distance, or unobservably sicker, may tend to live nearer to or farther from OB hospitals, biasing estimates related to distance. Second, mothers who are unobservably more sensitive to any hospital characteristic k , or unobservably sicker, may tend to live nearer to or farther from OB hospitals with characteristic k , biasing estimates related to k .

I examine plausible channels through which home location may relate to unobserved preference or underlying health shocks. As proxies for these channels, I use mothers' characteristics not included in estimation: marital status, high school completion, hepatitis C diagnosis, height, and a measure of environmental exposure, outdoor nitrogen dioxide (NO₂) concentration by census block group.¹⁵ To summarize a mother's home location, I consider the following attributes: distance to the nearest OB hospital in the mother's choice set, whether that nearest option is UAMS, whether it has a NICU, and its logged non-OB beds. Using ordinary least squares, I regress each attribute on each proxy, controlling for the mothers' characteristics that appear in either the demand model (Equations 10 and 11) or health model (Equation 12). Then, I report the coefficient on the proxy. Table D1 shows the results.

All coefficients are close to zero and statistically insignificant. This rules out the most

¹⁵Data on NO₂ concentration in 2020 come from the Environmental Protection Agency's Environmental Justice Screening and Mapping Tool. I impute each mother's expected value based on her county and age.

Table B4: Hospital Summary Statistics

	OB Unit	No OB Unit
Count		
UAMS (Teaching)	1	0
WCWH (Women's)	1	0
Public	4	5
Non-profit	25	26
For-profit	8	3
NICU	9	0
Mean		
Non-OB Beds	157.5	53.0
Patient Reviews (1 SD)	-0.38	0.41
Patient Outcomes (1 SD)	-0.29	0.31
Wage Index (1 SD)	0.3	-0.3
Observations	37	34

Sources: AHA, CMS. Patient reviews: composite of 10 questions in Medicare patient survey, normalized to mean 0 and SD 1. Patient outcomes: composite of Medicare mortality and readmission rates for pneumonia, COPD, and heart failure, normalized to mean 0 and SD 1. Wage index: regional hospital worker wage, normalized to mean 0 and SD 1.

plausible channels through which mothers' home locations may relate to unobserved shocks, suggesting that any variation in mothers' home locations that remains after conditioning on observed characteristics is driven by factors unrelated to OB care.

Table C1: Insurers Available to Each Mother

Payer Type	Births	Insurers
Medicaid	10526	1
ARHOME	4776	23
Employer-Sponsored	2406	35 (avg 1.3 in group)
State Employee	1262	7
ACA Marketplace	1117	19
Total	20087	85

Sample: births with claims data. Assumptions: mothers only choose within payer type, and mothers with employer-sponsored insurance only choose within group.

Table C2: Insurers Are Chosen Before, and Not During, Pregnancy

	Commercial	Medicaid
Had Birth Insurer Before Pregnancy (%)	73.6	61.7
Any Other Insurer During Pregnancy (%)	28.7	9.1
With Higher OOP (%)	3.4	5.2
With Equal OOP (%)	19.1	4.2
With Lower OOP (%)	7.1	0.0
Observations	2216	2368

Sample: births with claims data in Oct 9–Dec 31, excluding outside option births. Pregnancy length: 280 days. Higher OOP: other insurer has higher average OOP at chosen OB hospital than birth insurer, or chosen OB hospital is out-of-network for other insurer. Equal OOP: other insurer has equal average OOP as birth insurer. Lower OOP: other insurer has lower average OOP than birth insurer.

E Recovering Intensity, Price, and Out-of-Pocket Spending

For mother i with insurer m , I observe her payment and out-of-pocket spending (split among copay, deductible, and coinsurance). I assume payment and out-of-pocket spending are

$$\text{Payment}_{ij} = r_i p_{mj} \quad (\text{E1})$$

$$\text{OOP}_{ij} = \text{Copay}_i + \text{Deductible}_i + \text{CoinsuranceRate}_i (\text{Payment}_{ij} - \text{Copay}_i - \text{Deductible}_i) \quad (\text{E2})$$

where r_i is resource use intensity and p_{mj} is price per unit of intensity.

To recover r_i and p_{mj} , I estimate

$$\ln \text{Payment}_{ij} = \rho' \mathbf{x}_i + \text{FE}_{mj} + v_{ij}, \quad E[v_{ij}] = 0 \quad (\text{E3})$$

where $\mathbf{x}_i$ consist of a constant, mother's age, mother's race, mother's BMI, a diabetes indicator, a hypertension indicator, a smoker indicator, a prior birth indicator, a prior caesarean section indicator, logged home county population density, and home county population shares by income bin. For intuition, the estimate of r_i is approximately $\exp(\rho' \mathbf{x}_i)$, and the estimate of p_{mj} is approximately $\exp(\widehat{\text{FE}}_{mj})$.

To turn those approximations into unbiased estimates, I make two adjustments. The first adjustment is to apply empirical Bayes shrinkage to $\widehat{\text{FE}}_{mj}$. To do so, I estimate

$$\text{FE}_{mj} = \text{FE}_m + \text{FE}_j + \tilde{v}_{mj}, \quad E[\tilde{v}_{mj}] = 0 \quad (\text{E4})$$

where FE_{mj} , FE_m , and FE_j are estimated with normal error. I obtain shrunk versions of $\widehat{\text{FE}}_m$ and $\widehat{\text{FE}}_j$. Then, $\widehat{\text{FE}}_{mj}$ is shrunk toward $\widehat{\text{FE}}_m + \widehat{\text{FE}}_j$. If insurer-hospital pair mj has no births with

Table D1: Ruling Out Ways Home Location May Relate to Unobserved Shocks

Proxy for Unobserved Shocks	Attribute of Nearest Option			
	Distance (mi)	UAMS	NICU	ln(Non-OB Beds)
Conditional on $\mathbf{x}_i^{\text{demand}}$				
Married	0.063 (0.647)	-0.071 (0.072)	-0.058 (0.045)	0.006 (0.085)
High School	-0.249 (0.334)	0.027 (0.034)	-0.005 (0.035)	0.009 (0.036)
Hepatitis C	-0.584 (0.729)	-0.056 (0.073)	0.054 (0.086)	0.088 (0.100)
Height (per SD, 2.8 in)	-0.023 (0.101)	0.010 (0.012)	-0.003 (0.011)	0.010 (0.012)
NO ₂ (per SD, 1.5 ppb)	0.035 (1.735)	0.190 (0.201)	0.167 (0.121)	0.119 (0.214)
Conditional on $\mathbf{x}_i^{\text{health}}$				
Married	-0.069 (0.169)	-0.001 (0.006)	-0.002 (0.013)	0.012 (0.017)
High School	-0.206 (0.264)	-0.003 (0.011)	0.005 (0.017)	0.020 (0.022)
Hepatitis C	-0.119 (0.684)	-0.042 (0.057)	0.017 (0.049)	-0.026 (0.055)
Height (per SD, 2.8 in)	0.030 (0.055)	0.001 (0.002)	-0.001 (0.003)	0.004 (0.006)
NO ₂ (per SD, 1.5 ppb)	0.225 (1.545)	0.122 (0.139)	0.041 (0.110)	0.029 (0.156)
Observations	19843			

Sample: births with claims data, excluding outside option births. Attributes of nearest options are regressed with OLS on proxies for unobserved preference or underlying health shocks, controlling for mothers' characteristics that appear in either the demand or health model. Coefficients on proxies are reported. County-clustered SEs.

claims data, then $\widehat{\text{FE}}_{mj} = \widehat{\text{FE}}_m + \widehat{\text{FE}}_j$ exactly. As a note, the unshrunk prices are maintained in one place: I subtract markups from unshrunk prices to obtain implied marginal costs when estimating commercial price-setting parameters.

The second adjustment is to correct the estimates of r_i and p_{mj} to account for Jensen's inequality because estimation errors in $\hat{\rho}$ and $\widehat{\text{FE}}_{mj}$ enter non-linearly.

Finally, I multiply r_i by a constant, and divide p_{mj} by the same constant, so the average of r_i is 1. Then, p_{mj} can be interpreted as the price of an average birth.

F Additional Results: Demand and Health

The following tables present additional results from estimation of the model of OB hospital demand and maternal and infant health. Table F1 shows the effects of distance. Table F2 shows the effects of hospital size and having a NICU. Table F3 shows the effects of UAMS (the only hospital with an obstetrics and gynecology residency program), WCWH (the only women's hospital), and the outside option. Table F4 shows the demand effects of the patient reviews index and the health effects of the patient outcomes index. Table F5 shows the health effects of mothers' characteristics. Table F6 shows the joint distributions of unobserved shocks.

Table F1: Demand and Health Results: Distance

	Median Δ Utility (\$)	Avg Δ Pr(Healthy) (%)
Distance (mi)	−225.8 (69.7)	−0.0213 (0.0092)
Δ Effect of Distance per Δ Distance	−18.8 (12.2)	
Δ Effect of Distance (mi) per Δx_i :		
Age (yr)	0.1 (1.8)	−0.0006 (0.0007)
BMI	1.8 (1.4)	0.0010 (0.0005)
Prior Births>0	−42.9 (25.3)	0.0100 (0.0098)
Prior C-sections>0	40.1 (24.3)	0.0001 (0.0138)
Medicaid	−109.0 (41.6)	
ARHOME	−85.7 (35.1)	
2×Pop Density	−70.6 (34.6)	
	Avg Δ Utility (\$)	
Distance (mi)	−940.3 (522.4)	
Observations	20087	

Sample: births with claims data. Utility effects are median marginal effects, except last row which is average marginal effect. Health effects are average marginal effects. Utility effects evaluated at chosen distances. County-clustered SEs for estimates related to mother-level variation, model-based SEs for estimates related to only hospital-level variation.

G Additional Results: Effects of OB Provision

The following figures present additional results related to the effects of OB provision. Figure G1 shows heterogeneity in access-to-care and maternal and infant health effects. It compares effects on rural and urban mothers, as well as effects on Medicaid and commercial mothers. Figure G2 shows the distance effects of OB provision. Figure G3 shows the role of distance in the access-to-care and maternal and infant health effects. Figure G4 compares the net externalities of rural and urban OB hospitals.

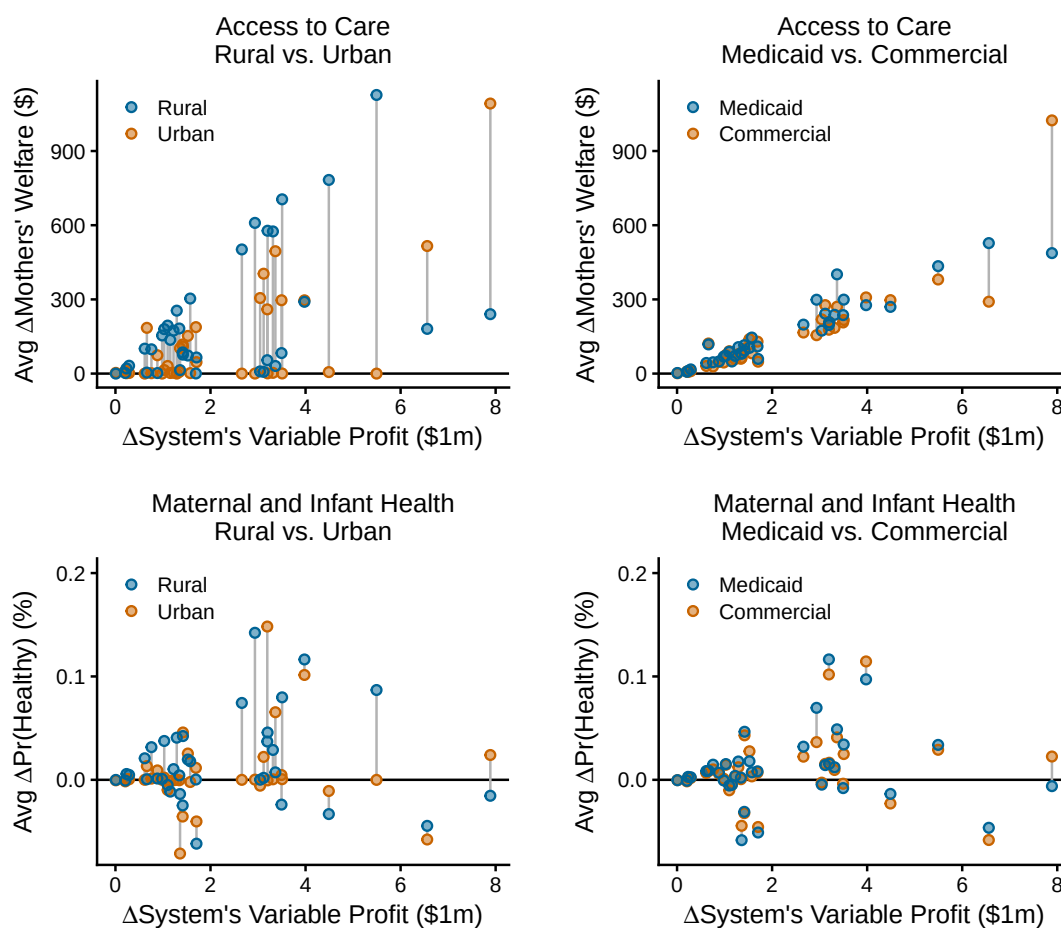


Figure G1: Heterogeneity in Access-to-Care and Maternal and Infant Health Effects Across Types of Mothers

Sample: OB hospitals. Horizontal axis is change in system's expected variable profit caused by hospital's OB provision. Vertical axes are components of variable social surplus, each divided by births in group: change in mothers' demand-based welfare before OOP, and value of change in expected no-adverse-outcome births caused by hospital's OB provision.

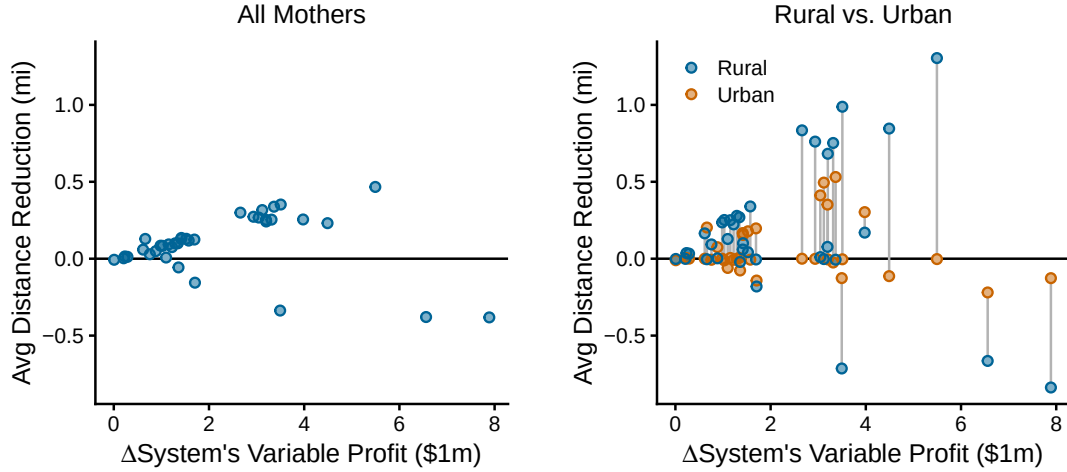


Figure G2: Distance Effects of OB Provision

Sample: OB hospitals. Horizontal axis is change in system's expected variable profit caused by hospital's OB provision. Vertical axis is reduction in expected travel distance, divided by births.

H Stylized Dynamic Model of OB Provision

In Section 6, OB provision is modeled as a static decision: it depends on current-period profitability, ignoring past or future outcomes. A realistic dynamic model would differ in two ways. First, opening an OB unit may require a one-time sunk cost (e.g., construction). A hospital may refrain from operating an OB unit because the sunk cost is too high, even if it would earn a positive per-period profit. Second, hospitals may be forward-looking. A hospital may operate an unprofitable OB unit, or not operate a potentially profitable OB unit, due to expectations about the future.

These features may lead to violations of the static OB provision condition used for fixed cost estimation. If a hospital operates an OB unit that is unprofitable in the current period, I understate its fixed cost. If a hospital does not operate an OB unit that would be profitable in the current period, I overstate its fixed cost. In this section, I present a stylized dynamic model to illustrate assumptions under which these violations do not happen, and I argue that they are reasonable in this setting.

Stylized Dynamic Model Suppose time periods are indexed by $t \in \{0, 1, 2, \dots\}$, and future payoffs are discounted at factor $\chi \in (0, 1)$. In period t , an OB unit at hospital j generates variable profit π_{jt} for its system. Operating an OB unit costs fixed cost FC_j every period. Opening a new OB unit costs a one-time sunk cost $SC_j \leq \overline{SC}$, and closing an OB unit is free. Hospital j makes OB provision decision $\text{Prov}_{jt} \in \{0, 1\}$ at every period t .

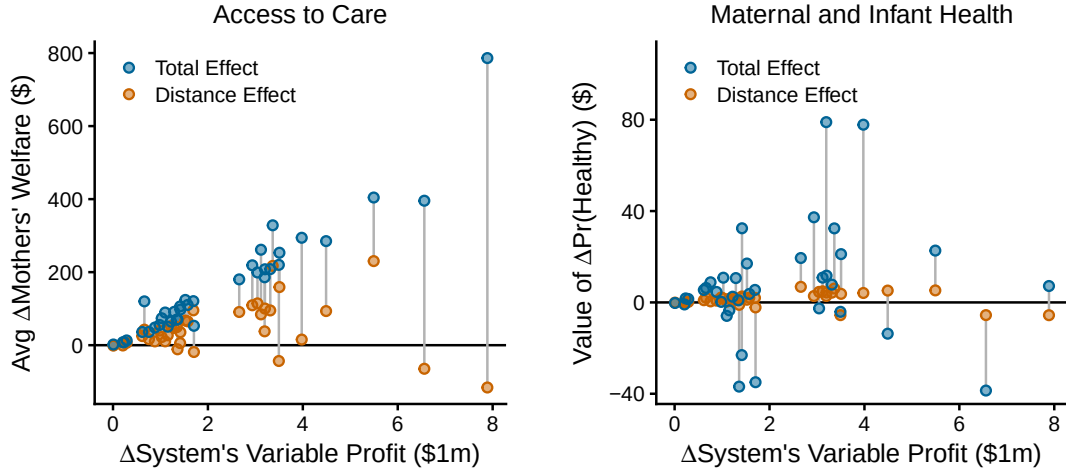


Figure G3: Access-to-Care and Maternal and Infant Health Effects, Separating Distance Effect From Total Effect

Sample: OB hospitals. Horizontal axis is change in system's expected variable profit caused by hospital's OB provision. Vertical axes are components of variable social surplus, each divided by Arkansas births: change in mothers' demand-based welfare before OOP, and value of change in expected no-adverse-outcome births caused by hospital's OB provision. Distance effect is explained only by reduction in expected travel distance.

A key assumption is that variable profits decline steadily: $\pi_{j,t+1} = \pi_{jt} - g_j$ with $g_j > 0$ for all hospitals j and periods t . Then, there is no reason to delay opening an OB unit, so a hospital either opens an OB unit at period 0 or never opens one. After opening, an OB unit operates until its variable profit falls below its fixed cost, at which point it closes and never reopens.

Suppose the market is observed at period $T > 0$. There are three types of hospitals. First, hospital j may operate an OB unit. This implies

$$\pi_{jT} \geq FC_j \quad (H1)$$

Second, hospital j may have opened an OB unit at period 0 but since closed it. The fact that it closed at or earlier than period T implies

$$\pi_{jT} \leq FC_j \quad (H2)$$

Third, hospital j may have never opened an OB unit. The fact that it did not open one at period 0 implies no possible sequence of decisions would make the sunk cost worth it. In particular, keeping the OB unit open until period T would not be worth it, so

$$\sum_{t=0}^T \chi^t (\pi_{jT} - FC_j + g_j (T - t)) \leq SC_j \quad (H3)$$

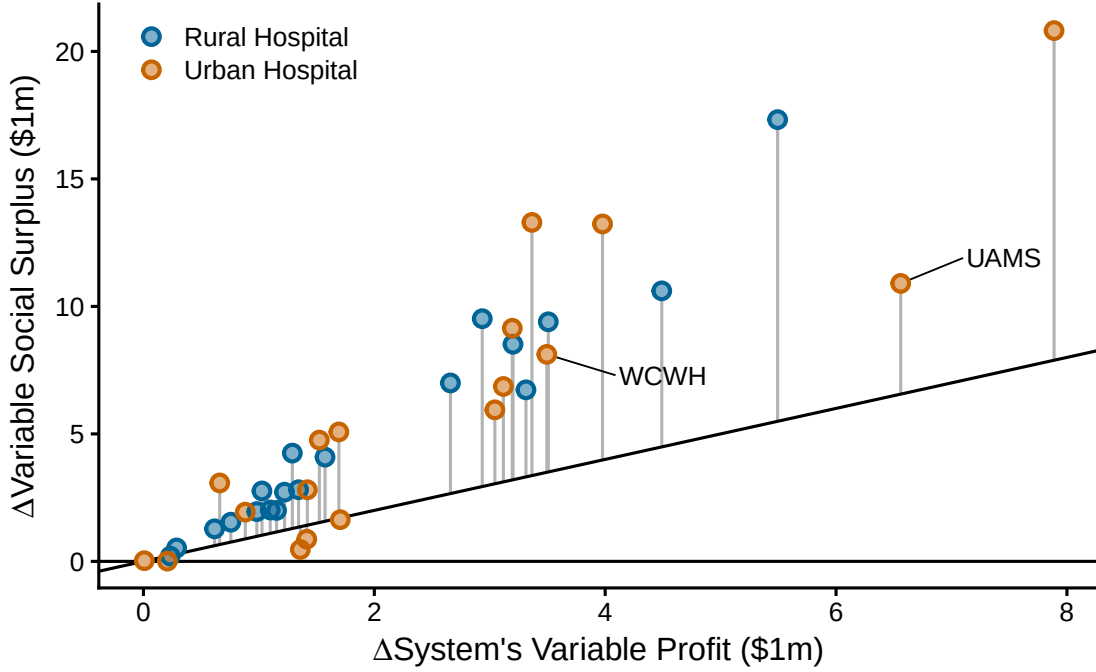


Figure G4: Net Externality of OB Provision, Rural vs. Urban Hospitals

Sample: OB hospitals. Horizontal axis is change in system's expected variable profit caused by hospital's OB provision. Vertical axis is variable social surplus, summing the following: change in mothers' demand-based welfare before OOP, expected variable cost savings, and value of change in expected no-adverse-outcome births caused by hospital's OB provision. Sloped line is 45° line, and vertical lines denote net externalities. Labeled hospitals are UAMS (teaching) and WCWH (women's).

which implies

$$\pi_{jT} - FC_j \leq \frac{SC_j - g_j \sum_{t=0}^T \chi^t (T - t)}{\sum_{t=0}^T \chi^t} \quad (\text{H4})$$

Because sunk cost SC_j is bounded above, T can be large enough that the right-hand side of Equation H4 is non-positive for all such hospitals j .

Thus, there exists a T' such that for all periods $T \geq T'$, all hospitals j satisfy

$$\text{Prov}_{jT} = \begin{cases} 1 & \text{if } \pi_{jT} > FC_j \\ 0 & \text{if } \pi_{jT} < FC_j \end{cases} \quad (\text{H5})$$

This is exactly the static OB provision condition.

Discussion of Stylized Dynamic Model The stylized model illustrates two assumptions under which a static model captures all relevant features of OB provision. The first is that the profitability of OB provision is declining steadily, and the second is that it has been declining for a long time. Both assumptions are reasonable in this setting given the history of OB provision in

Arkansas, as shown in Figure J1. Between 1995 and 2024, there are zero openings of OB units and 24 closures of OB units in existing hospitals. This is consistent with the stylized model's prediction that a hospital either opens an OB unit at its first opportunity or never opens one, and an OB unit closes and never reopens after its per-period profit turns negative.

I Out-of-Sample Check Using Actual OB Unit Closures

Figure I1 highlights, in this setting, two hospitals that closed their OB units in 2020–2021 and four hospitals that closed their OB units in 2023–2024. In the figure, each hospital is ranked according to the expected change in its system's variable profit if it switched from non-provision to provision, before demand-based quality and marginal cost shocks are realized. The higher this measure is, the more likely a hospital is to operate an OB unit, and this relationship is used to estimate fixed cost.

The hospitals that actually closed their OB units all lie in the lower half of the ranking. In particular, the hospitals that closed their OB units in 2023–2024 are the least, third-least, fifth-least, and sixth-least profitable OB hospitals in 2022. This out-of-sample finding is consistent with the model's prediction.

J History of OB Provision in Arkansas

Figure J1 shows the history of OB provision in this setting in 1995–2024. There are few hospital openings and closures: 6 openings and 7 closures. There are also few positive changes in OB provision: 2 openings of new OB hospitals and zero additional OB units in existing hospitals. Most frequent are negative changes in OB provision: 2 closures of old OB hospitals and 24 closed OB units in existing hospitals. In such a setting where OB unit closures are frequent and OB unit openings are not, the effect of OB provision is equivalent to the effect of preventing OB unit closures.

K Additional Results: Policy Evaluation

The following tables present additional results related to evaluation of the Healthy Moms, Healthy Babies Act's Medicaid price increase. Table K1 decomposes the policy's effects on the individual components of social welfare. Tables K2 and K3 show heterogeneity in the policy's access-to-care and maternal and infant health effects. They compare effects on rural and urban mothers, as well as effects on Medicaid and commercial mothers. Table K4 shows the policy's

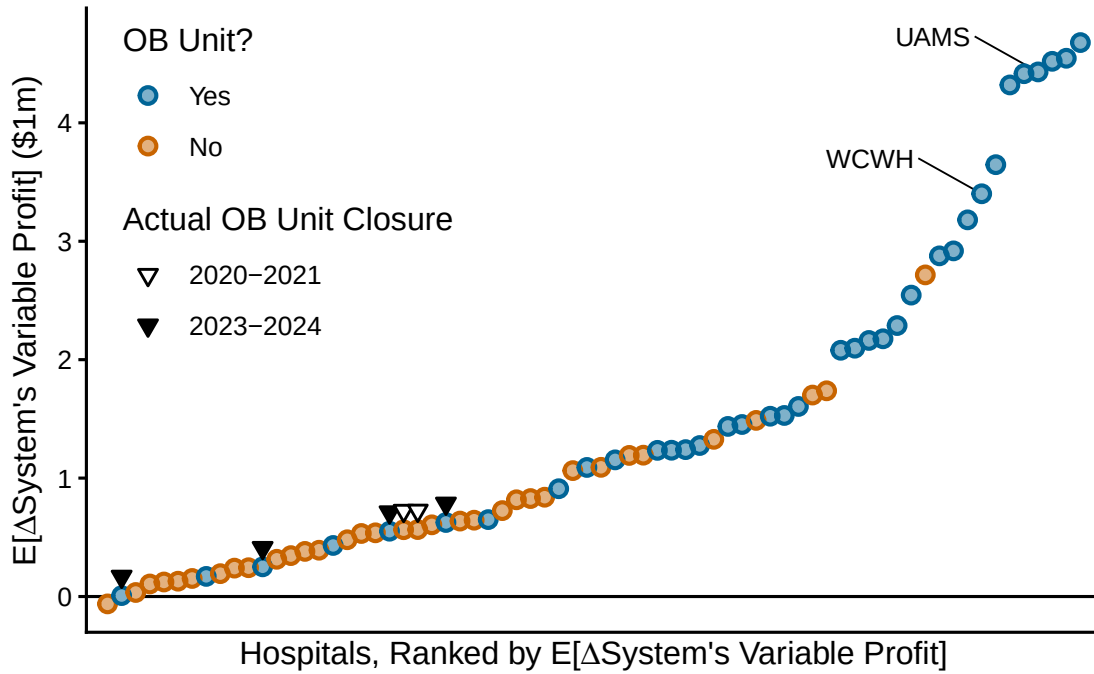


Figure I1: Actual OB Unit Closures and Predicted Profitability

Sample: all hospitals. Source for OB unit closures: UAMS. Vertical axis is change in system's expected variable profit caused by hospital's OB provision, before demand-based quality and marginal cost shocks are realized. Horizontal axis is ranked low to high. Labeled hospitals are UAMS (teaching) and WCWH (women's). One outlier is omitted: a hospital with \$14.88M.

distance effects. Table K5 shows the role of distance in the policy's access-to-care and maternal and infant health effects. Table K6 shows the policy's effects on rural OB hospitals, urban OB hospitals, and Medicaid spending.

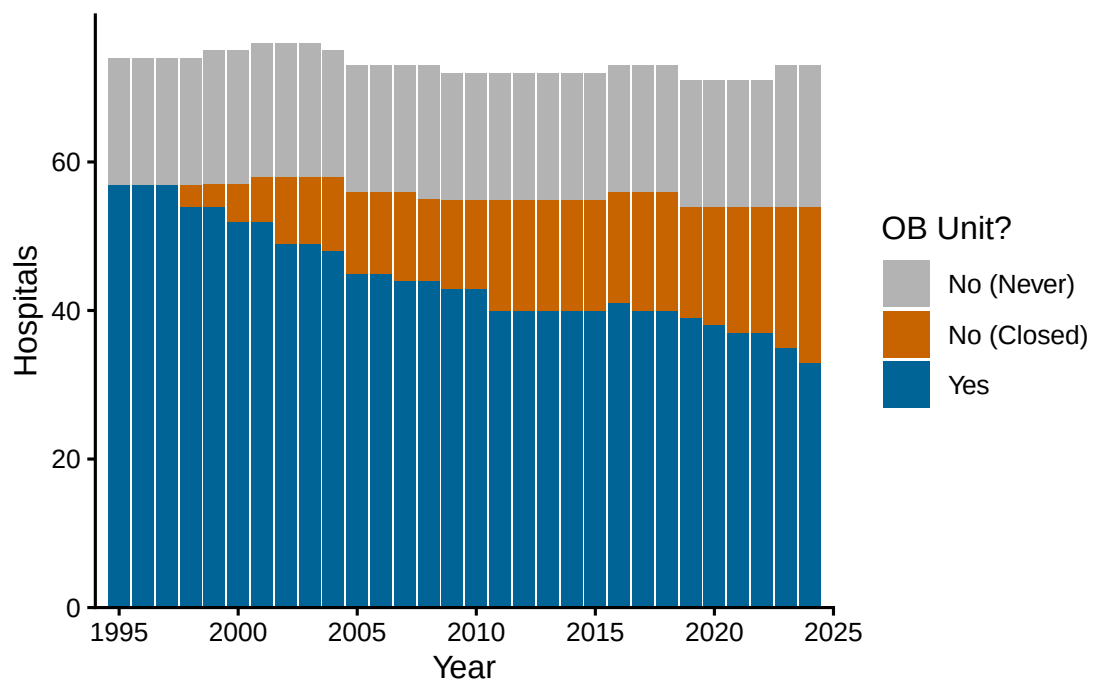


Figure J1: History of OB Provision in Arkansas

Data source: AHA. Sample: hospitals in Arkansas (except Crittenden/Miller/Little River Counties and specialty/VA hospitals).

Table F2: Demand and Health Results: Hospital Size and NICU

	Avg Δ Utility (\$)	Avg Δ Pr(Healthy) (%)
2×Non-OB Beds	3486.4 (1862.3)	−0.190 (0.400)
Δ Effect of 2×Non-OB Beds per Δx_i :		
Age (yr)	119.8 (29.2)	−0.001 (0.014)
BMI	57.3 (15.4)	0.023 (0.013)
Prior Births>0	−506.3 (148.3)	0.166 (0.178)
Prior C-sections>0	633.7 (374.8)	0.130 (0.287)
Medicaid	−238.4 (392.4)	
ARHOME	−152.5 (362.6)	
2×Pop Density	351.0 (121.9)	
NICU	−855.6 (2669.8)	0.761 (1.128)
Δ Effect of NICU per Δx_i :		
Age (yr)	−33.5 (28.6)	0.068 (0.060)
BMI	−70.7 (33.9)	0.005 (0.033)
Prior Births>0	−193.5 (282.5)	−0.324 (0.760)
Prior C-sections>0	−1004.1 (645.2)	−0.296 (0.868)
Medicaid	−1132.8 (815.7)	
ARHOME	−1263.5 (656.3)	
2×Pop Density	90.2 (200.5)	
Observations	20087	

Sample: births with claims data. Average marginal effects. County-clustered SEs for estimates related to mother-level variation, model-based SEs for estimates related to only hospital-level variation.

Table F3: Demand and Health Results: UAMS, WCWH, and Outside Option

	Avg Δ Utility (\$)	Avg Δ Pr(Healthy) (%)
UAMS	-2678.7 (5879.6)	0.243 (1.560)
Δ Effect of UAMS per Δx_i :		
Age (yr)	78.8 (42.0)	-0.085 (0.057)
BMI	62.2 (29.5)	-0.102 (0.040)
Prior Births>0	498.9 (426.4)	-1.500 (0.655)
Prior C-sections>0	-4304.8 (1635.6)	1.614 (1.207)
Medicaid	5175.7 (1828.6)	
ARHOME	2619.7 (926.6)	
2×Pop Density	323.9 (186.3)	
WCWH	9506.9 (3733.7)	
Outside Option	-24017.5 (5492.4)	-4.676 (2.497)
Observations	20087	

Sample: births with claims data. Average marginal effects. Effects relative to a hospital with same observed characteristics and zero quality shock. County-clustered SEs for estimates related to mother-level variation, model-based SEs for estimates related to only hospital-level variation.

Table F4: Demand and Health Results: Patient Reviews and Patient Outcomes

	Avg Δ Utility (\$)
Patient Reviews (1 SD)	−497.4 (1480.4)
Δ Effect of Patient Reviews (1 SD) per Δx_i :	
Age (yr)	30.5 (16.6)
BMI	40.7 (14.4)
Prior Births>0	−898.8 (306.7)
Prior C-sections>0	184.1 (225.9)
Medicaid	−1103.9 (603.3)
ARHOME	−1113.1 (610.3)
2×Pop Density	−184.2 (127.4)
	Avg Δ Pr(Healthy) (%)
Patient Outcomes (1 SD)	−0.206 (0.242)
Δ Effect of Patient Outcomes (1 SD) per Δx_i :	
Age (yr)	−0.004 (0.016)
BMI	0.004 (0.013)
Prior Births>0	0.521 (0.257)
Prior C-sections>0	0.108 (0.363)
Observations	20087

Sample: births with claims data. Average marginal effects. County-clustered SEs for estimates related to mother-level variation, model-based SEs for estimates related to only hospital-level variation.

Table F5: Demand and Health Results: Mothers' Characteristics in Health

	Avg Δ Pr(Healthy) (%)		Avg Δ Pr(Healthy) (%)
Age (yr)	−0.028 (0.033)	Race (vs. Asian):	
BMI	0.002 (0.016)	Black	−0.220 (0.875)
Prior Births>0	1.725 (0.349)	Native American	1.598 (1.340)
Prior C-sections>0	−0.521 (0.353)	Pacific Islander	0.535 (0.687)
Medicaid	0.067 (0.507)	Race Unknown	1.564 (0.776)
ARHOME	−0.042 (0.534)	White	1.208 (0.816)
2×Pop Density	−0.091 (0.127)	Income Share (%), by % of Poverty Line:	
Diabetes	−2.305 (0.865)	<50%	0.148 (0.089)
Hypertension	−0.071 (0.688)	50–99%	−0.192 (0.076)
Smoker	−0.842 (0.363)	100–124%	0.137 (0.113)
		125–149%	0.111 (0.109)
		150–184%	0.130 (0.105)
		185–199%	−0.294 (0.182)
Observations		20087	

Sample: births with claims data. Average marginal effects. County-clustered SEs.

Table F6: Demand and Health Results: Unobserved Shocks

Preferences	Δ Effect per SD of ν_i^k (\$)	$\text{Corr}(\nu_i^k, \epsilon_i)$
Distance (mi)	−837.5 (362.5)	0.06 (0.06)
UAMS	7682.8 (3060.5)	−0.34 (0.20)
NICU	6424.0 (2272.8)	−0.03 (0.11)
2×Non-OB Beds	2672.1 (904.6)	−0.03 (0.12)
Patient Reviews (1 SD)	4429.1 (1348.9)	−0.01 (0.07)
	SD of ξ_j (\$)	SD of λ_j (%)
Demand-Based and Health-Based Quality	2190.0 (459.4)	1.04 (0.24)
$\text{Corr}(\xi_j, \lambda_j)$		0.59 (0.32)
Observations	20087	

Sample: births with claims data. Effect of distance preference shock is median marginal effect. For distance, a bigger shock is negative preference. For other characteristics, a bigger shock is positive preference. County-clustered SEs for estimates related to mother-level variation, model-based SEs for estimates related to only hospital-level variation.

Table K1: Healthy Moms, Healthy Babies Act: Decomposition of Social Welfare Effects

	Δ Mothers' Welfare (\$1m)	Δ Value of Healthy Births (\$1m)	Variable Cost Savings (\$1m)	Fixed Cost Savings (\$1m)
10% Variable Profit Decline	—	—	—	—
+ Medicaid Price +10%	[+1.6, +7.8]	[+0.1, +1.3]	[0.0, +0.1]	[−1.9, −0.7]
+ Medicaid Price +35%	[+3.7, +17.9]	[+0.2, +1.8]	[+0.2, +1.5]	[−6.1, −1.6]
+ Medicaid Price +70%	[+3.9, +28.4]	[+0.2, +2.1]	[+0.2, +2.3]	[−11.4, −1.7]
20% Variable Profit Decline	—	—	—	—
+ Medicaid Price +10%	[+1.8, +4.3]	[−0.1, +0.3]	[−0.7, 0.0]	[−2.0, −0.7]
+ Medicaid Price +35%	[+5.5, +14.2]	[+0.3, +0.9]	[−2.1, +0.4]	[−5.6, −2.2]
+ Medicaid Price +70%	[+7.9, +26.8]	[+0.5, +2.0]	[+0.3, +1.5]	[−9.9, −3.2]
30% Variable Profit Decline	—	—	—	—
+ Medicaid Price +10%	[+2.1, +4.2]	[0.0, +0.2]	[−0.1, 0.0]	[−1.4, −0.7]
+ Medicaid Price +35%	[+6.4, +14.9]	[+0.3, +0.7]	[−0.2, 0.0]	[−5.5, −2.3]
+ Medicaid Price +70%	[+10.5, +25.5]	[+0.6, +1.4]	[−1.4, +0.6]	[−10.0, −3.9]

Variable profits decline due to higher marginal costs. Actual policy is 70% Medicaid price increase. Reported values are predicted outcomes: averages over fixed cost shocks drawn from uniform posterior. I report the range of predicted outcomes across the estimated set of expected fixed cost. Outcomes are mothers' demand-based welfare before OOP, value of expected no-adverse-outcome births, expected variable cost savings, and fixed cost savings.

Table K2: Healthy Moms, Healthy Babies Act: Heterogeneity in Access-to-Care Effects Across Types of Mothers

	Avg Δ Mothers' Welfare (\$)			
	Rural	Urban	Medicaid	Commercial
10% Variable Profit Decline	—	—	—	—
+ Medicaid Price +10%	[+61, +415]	[+42, +146]	[+54, +330]	[+45, +173]
+ Medicaid Price +35%	[+167, +891]	[+86, +367]	[+127, +675]	[+105, +460]
+ Medicaid Price +70%	[+180, +1741]	[+86, +398]	[+132, +1002]	[+110, +784]
20% Variable Profit Decline	—	—	—	—
+ Medicaid Price +10%	[+62, +167]	[+52, +113]	[+60, +148]	[+52, +120]
+ Medicaid Price +35%	[+214, +592]	[+146, +355]	[+184, +491]	[+160, +400]
+ Medicaid Price +70%	[+354, +1313]	[+182, +557]	[+268, +978]	[+225, +709]
30% Variable Profit Decline	—	—	—	—
+ Medicaid Price +10%	[+70, +124]	[+61, +145]	[+69, +141]	[+61, +124]
+ Medicaid Price +35%	[+233, +481]	[+181, +519]	[+214, +485]	[+188, +443]
+ Medicaid Price +70%	[+436, +1078]	[+265, +669]	[+354, +893]	[+305, +706]

Variable profits decline due to higher marginal costs. Actual policy is 70% Medicaid price increase. Reported values are predicted outcomes: averages over fixed cost shocks drawn from uniform posterior. I report the range of predicted outcomes across the estimated set of expected fixed cost. Outcome is mothers' demand-based welfare before OOP, divided by births in group.

Table K3: Healthy Moms, Healthy Babies Act: Heterogeneity in Maternal and Infant Health Effects Across Types of Mothers

	Avg Δ Pr(Healthy) (%)			
	Rural	Urban	Medicaid	Commercial
10% Variable Profit Decline	—	—	—	—
+ Medicaid Price +10%	[+0.006, +0.121]	[+0.002, +0.021]	[+0.004, +0.070]	[+0.003, +0.047]
+ Medicaid Price +35%	[+0.017, +0.154]	[+0.005, +0.033]	[+0.011, +0.093]	[+0.008, +0.063]
+ Medicaid Price +70%	[+0.018, +0.204]	[+0.005, +0.023]	[+0.011, +0.107]	[+0.008, +0.072]
20% Variable Profit Decline	—	—	—	—
+ Medicaid Price +10%	[+0.001, +0.020]	[−0.006, +0.010]	[−0.002, +0.014]	[−0.004, +0.011]
+ Medicaid Price +35%	[+0.020, +0.065]	[+0.008, +0.025]	[+0.015, +0.046]	[+0.012, +0.035]
+ Medicaid Price +70%	[+0.034, +0.185]	[+0.011, +0.032]	[+0.023, +0.108]	[+0.016, +0.070]
30% Variable Profit Decline	—	—	—	—
+ Medicaid Price +10%	[+0.003, +0.011]	[+0.001, +0.005]	[+0.002, +0.008]	[+0.001, +0.007]
+ Medicaid Price +35%	[+0.014, +0.045]	[+0.012, +0.021]	[+0.014, +0.033]	[+0.013, +0.027]
+ Medicaid Price +70%	[+0.040, +0.108]	[+0.016, +0.032]	[+0.028, +0.071]	[+0.022, +0.051]

Variable profits decline due to higher marginal costs. Actual policy is 70% Medicaid price increase. Reported values are predicted outcomes: averages over fixed cost shocks drawn from uniform posterior. I report the range of predicted outcomes across the estimated set of expected fixed cost. Outcome is expected no-adverse-outcome births, divided by births in group.

Table K4: Healthy Moms, Healthy Babies Act: Distance Effects

	Avg Distance Reduction (mi)		
	All Mothers	Rural	Urban
10% Variable Profit Decline	—	—	—
+ Medicaid Price +10%	[+0.05, +0.32]	[+0.07, +0.54]	[+0.04, +0.19]
+ Medicaid Price +35%	[+0.12, +0.65]	[+0.19, +1.09]	[+0.08, +0.40]
+ Medicaid Price +70%	[+0.12, +0.93]	[+0.21, +2.00]	[+0.07, +0.34]
20% Variable Profit Decline	—	—	—
+ Medicaid Price +10%	[+0.05, +0.12]	[+0.07, +0.17]	[+0.04, +0.10]
+ Medicaid Price +35%	[+0.17, +0.43]	[+0.24, +0.63]	[+0.13, +0.31]
+ Medicaid Price +70%	[+0.25, +0.85]	[+0.41, +1.48]	[+0.16, +0.50]
30% Variable Profit Decline	—	—	—
+ Medicaid Price +10%	[+0.06, +0.10]	[+0.04, +0.12]	[+0.06, +0.09]
+ Medicaid Price +35%	[+0.20, +0.36]	[+0.22, +0.49]	[+0.17, +0.39]
+ Medicaid Price +70%	[+0.33, +0.71]	[+0.48, +1.11]	[+0.24, +0.49]

Variable profits decline due to higher marginal costs. Actual policy is 70% Medicaid price increase. Reported values are predicted outcomes: averages over fixed cost shocks drawn from uniform posterior. I report the range of predicted outcomes across the estimated set of expected fixed cost. Outcome is reduction in expected travel distance, divided by births.

Table K5: Healthy Moms, Healthy Babies Act: Access-to-Care and Maternal and Infant Health Effects, Separating Distance Effect From Total Effect

	Avg ΔMothers' Welfare (\$)		Avg ΔPr(Healthy) (%)	
	Total Effect	Distance Effect	Total Effect	Distance Effect
10% Variable Profit Decline	—	—	—	—
+ Medicaid Price +10%	[+49, +243]	[+20, +89]	[+0.004, +0.057]	[+0.001, +0.005]
+ Medicaid Price +35%	[+115, +555]	[+54, +281]	[+0.009, +0.076]	[+0.002, +0.012]
+ Medicaid Price +70%	[+120, +880]	[+56, +409]	[+0.009, +0.088]	[+0.003, +0.019]
20% Variable Profit Decline	—	—	—	—
+ Medicaid Price +10%	[+56, +132]	[+17, +50]	[−0.003, +0.012]	[+0.001, +0.002]
+ Medicaid Price +35%	[+170, +440]	[+69, +183]	[+0.013, +0.040]	[+0.004, +0.009]
+ Medicaid Price +70%	[+244, +828]	[+114, +366]	[+0.019, +0.087]	[+0.005, +0.017]
30% Variable Profit Decline	—	—	—	—
+ Medicaid Price +10%	[+64, +131]	[+26, +40]	[+0.002, +0.007]	[+0.001, +0.002]
+ Medicaid Price +35%	[+200, +462]	[+81, +155]	[+0.014, +0.030]	[+0.004, +0.007]
+ Medicaid Price +70%	[+326, +789]	[+144, +314]	[+0.025, +0.060]	[+0.007, +0.015]

Variable profits decline due to higher marginal costs. Actual policy is 70% Medicaid price increase. Reported values are predicted outcomes: averages over fixed cost shocks drawn from uniform posterior. I report the range of predicted outcomes across the estimated set of expected fixed cost. Outcomes are mothers' demand-based welfare before OOP, and expected no-adverse-outcome births, each divided by births. Distance effect is explained only by reduction in expected travel distance.

Table K6: Healthy Moms, Healthy Babies Act: Rural OB Hospitals, Urban OB Hospitals, and Medicaid Spending

	Rural OB Hospitals	Urban OB Hospitals	ΔMedicaid Spending (\$1m)
10% Variable Profit Decline	[15.4, 18.3]	[16.0, 17.5]	—
+ Medicaid Price +10%	[15.6, 18.6]	[16.8, 17.8]	[+4.1, +4.2]
+ Medicaid Price +35%	[17.0, 19.0]	18.0	+14.6
+ Medicaid Price +70%	19.0	18.0	[+29.1, +29.2]
20% Variable Profit Decline	[14.2, 17.5]	[15.0, 16.9]	—
+ Medicaid Price +10%	[14.7, 17.8]	[15.6, 17.2]	+4.2
+ Medicaid Price +35%	[15.3, 18.4]	[16.9, 17.8]	[+14.5, +14.6]
+ Medicaid Price +70%	[16.8, 18.9]	18.0	[+29.1, +29.3]
30% Variable Profit Decline	[13.1, 16.8]	[13.8, 16.3]	—
+ Medicaid Price +10%	[13.4, 17.1]	[14.4, 16.6]	+4.2
+ Medicaid Price +35%	[13.9, 17.7]	[16.3, 17.4]	[+14.5, +14.6]
+ Medicaid Price +70%	[15.2, 18.5]	[16.9, 17.9]	[+29.1, +29.2]

Variable profits decline due to higher marginal costs. Actual policy is 70% Medicaid price increase. Reported values are predicted outcomes: averages over fixed cost shocks drawn from uniform posterior. I report the range of predicted outcomes across the estimated set of expected fixed cost. Outcomes are number of OB hospitals and Medicaid spending.